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(54) **TRANSPORTABLE MACHINE INCLUDING A TRACK SYSTEM**

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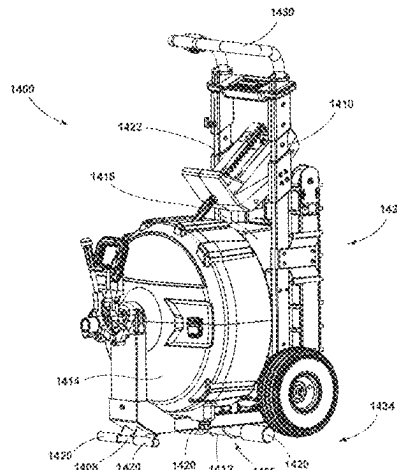
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(57)

ABSTRACT

A transportable machine including an elongated frame having a top end and a bottom end, a handle disposed on the top end of the frame, a base disposed on the bottom end of the frame, where the base and the frame together define a space configured to receive and support a load, and a wheel disposed on the bottom end of the frame. A track is coupled to the frame, where the track includes a roller and an endless belt configured to rotate about the roller. A motor is coupled

(Continued)



to the roller, where the motor is operable to drive the endless belt, and a power supply provides power to the motor.

14 Claims, 26 Drawing Sheets

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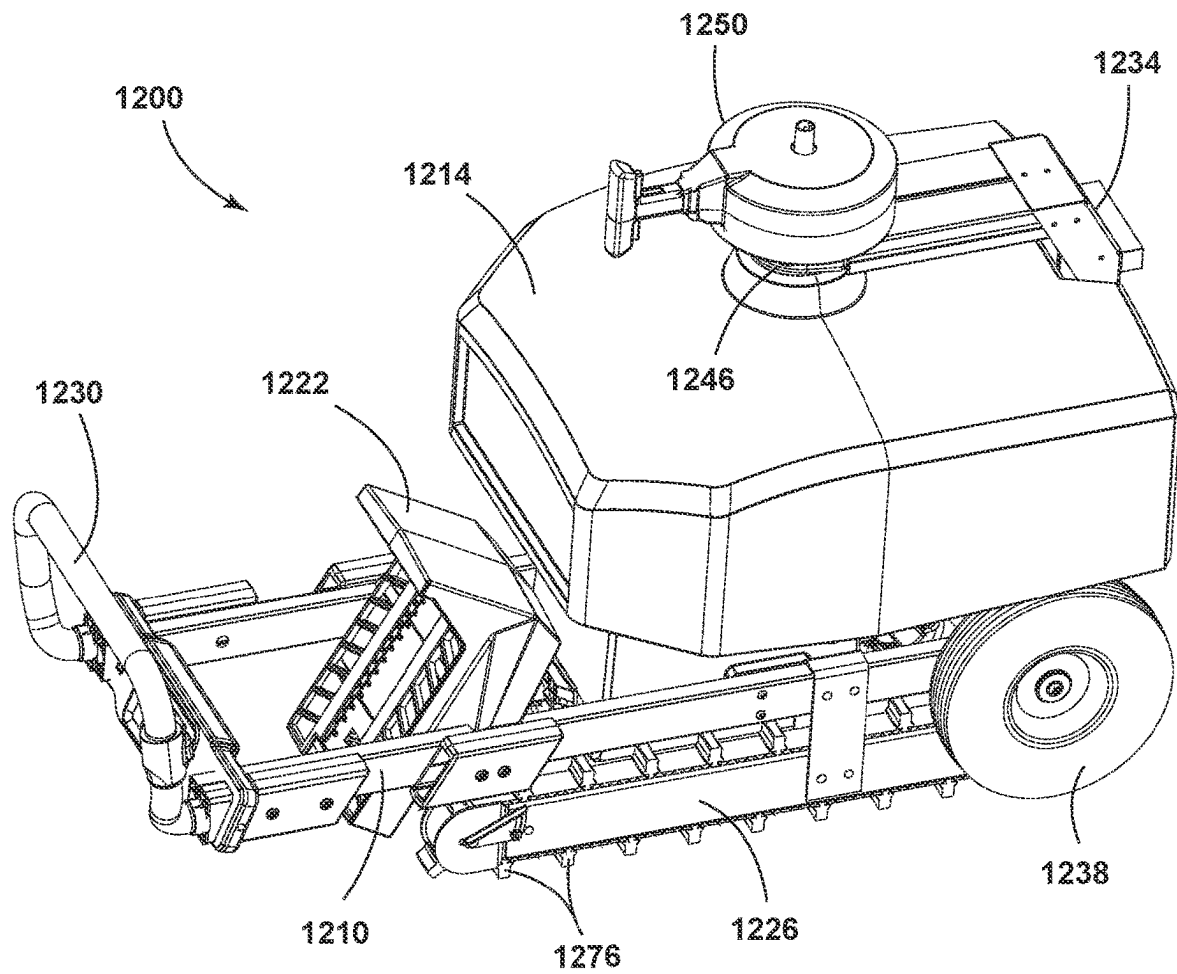
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**FIG. 1**

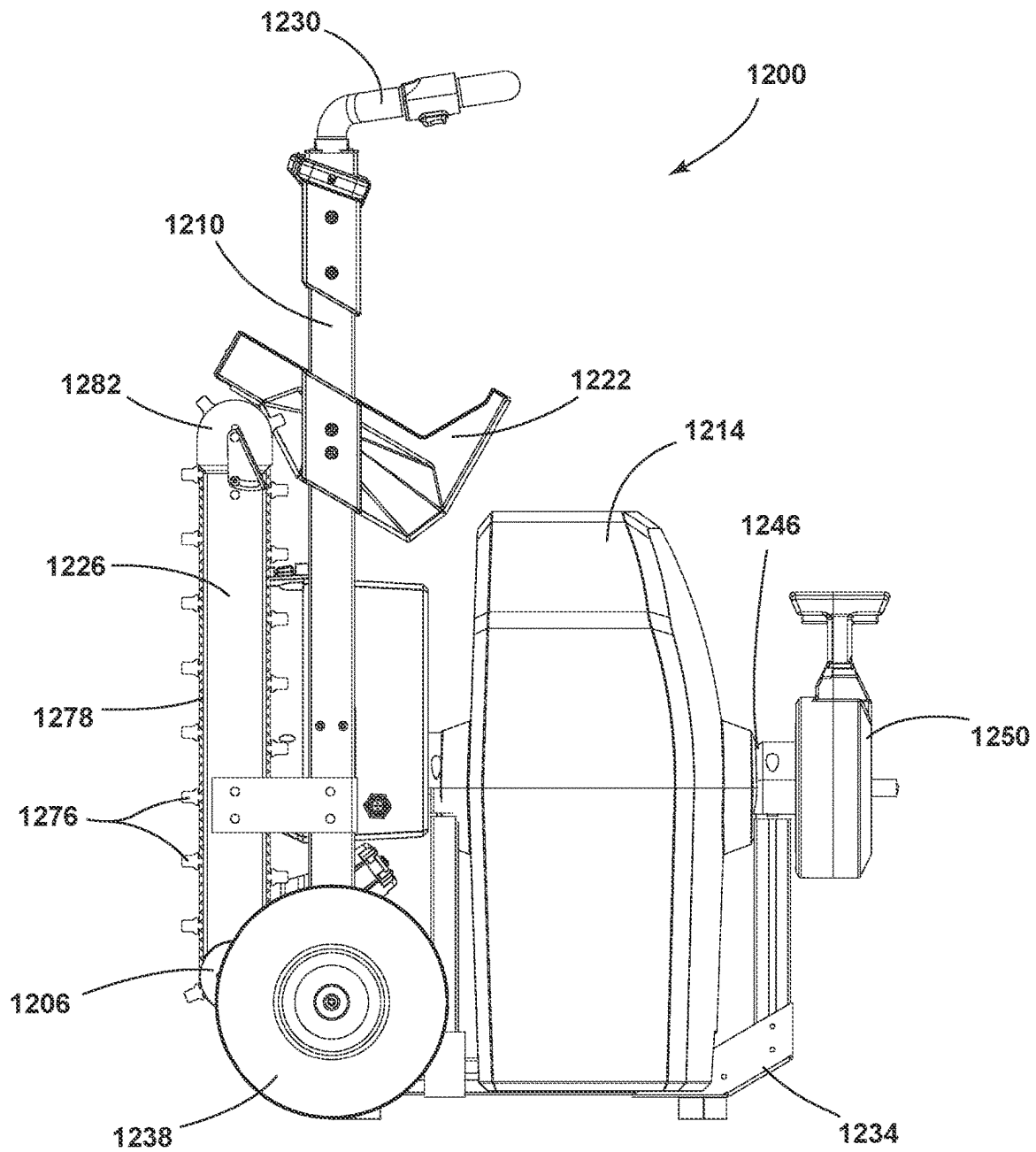


FIG. 2

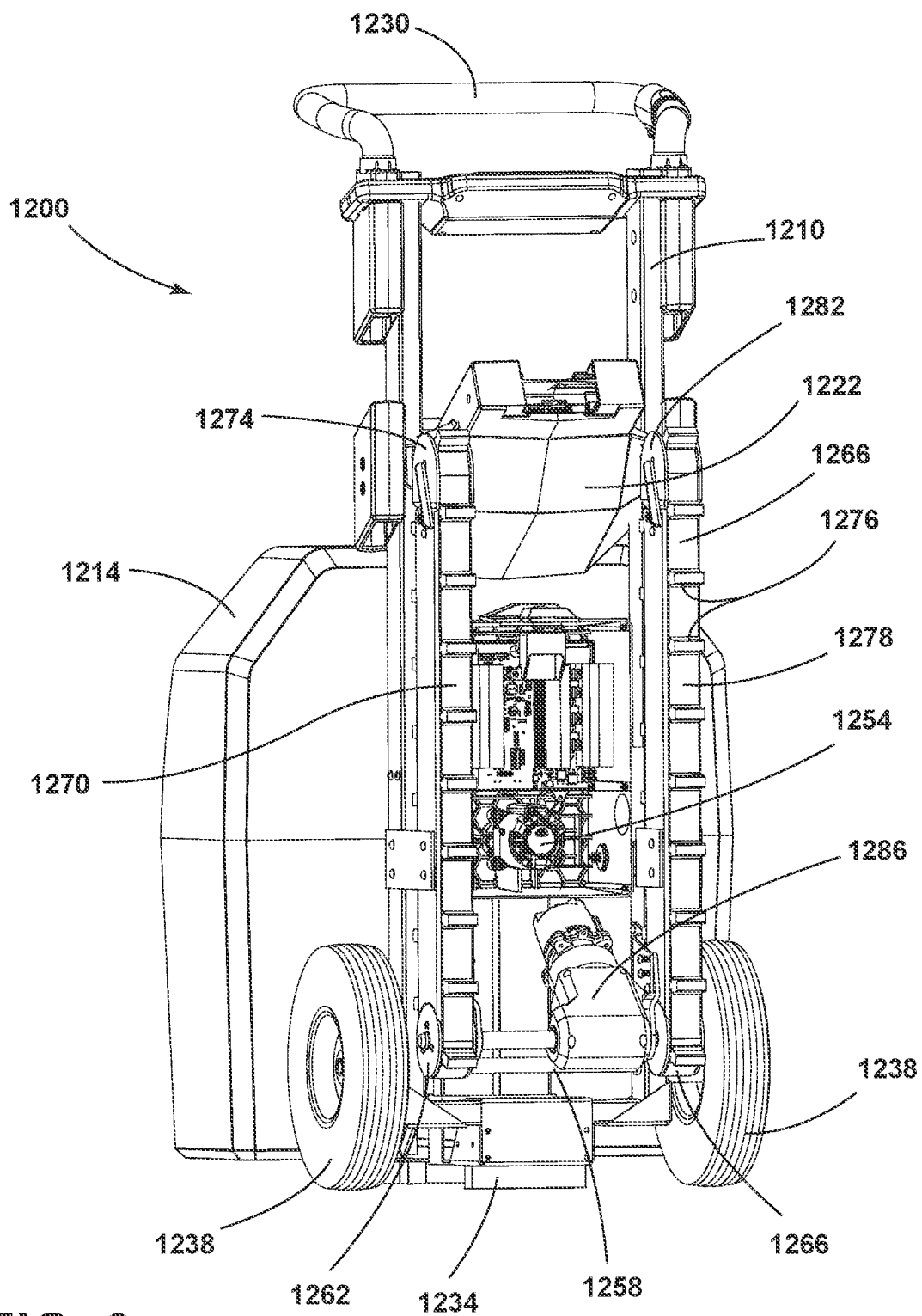


FIG. 3

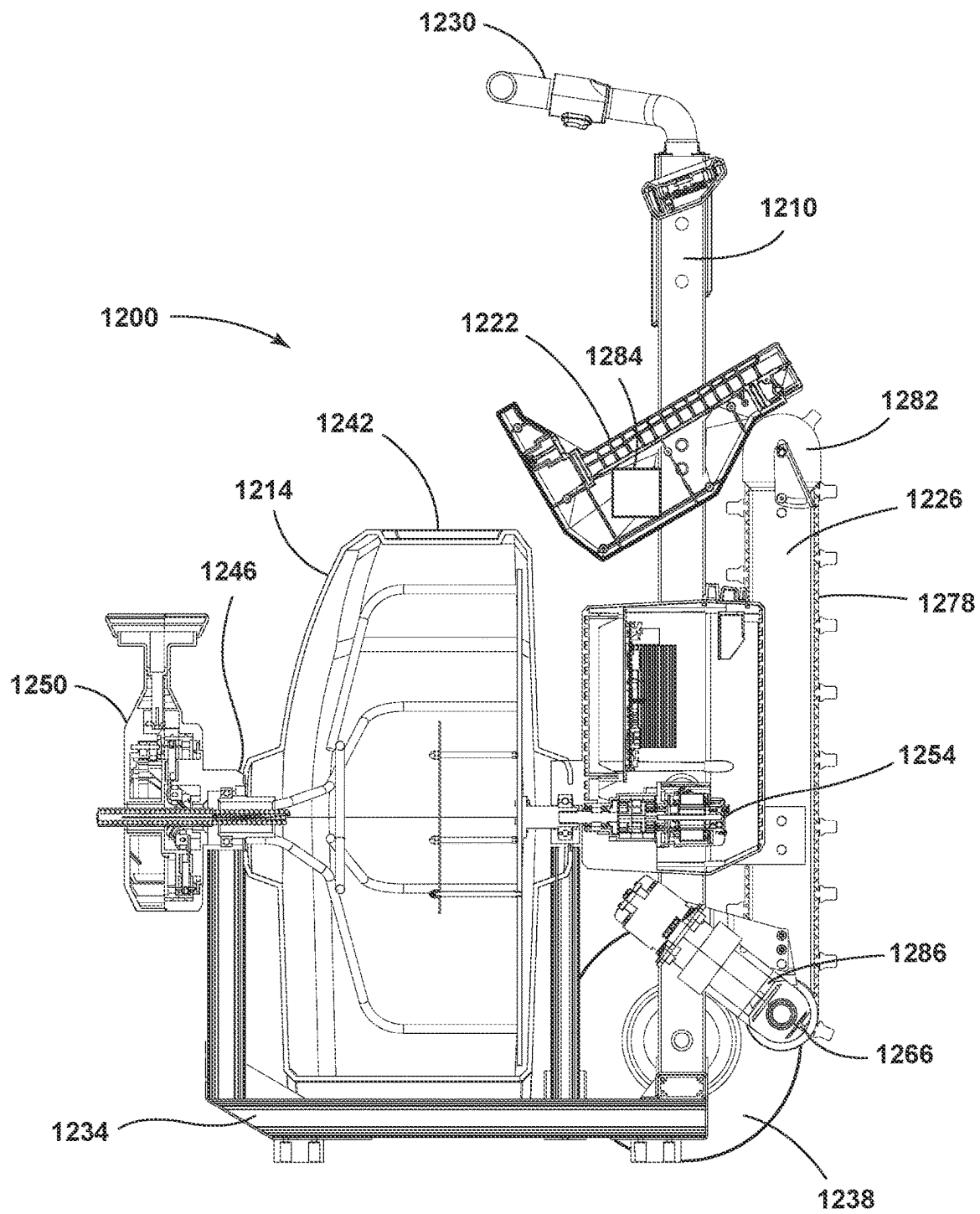


FIG. 4

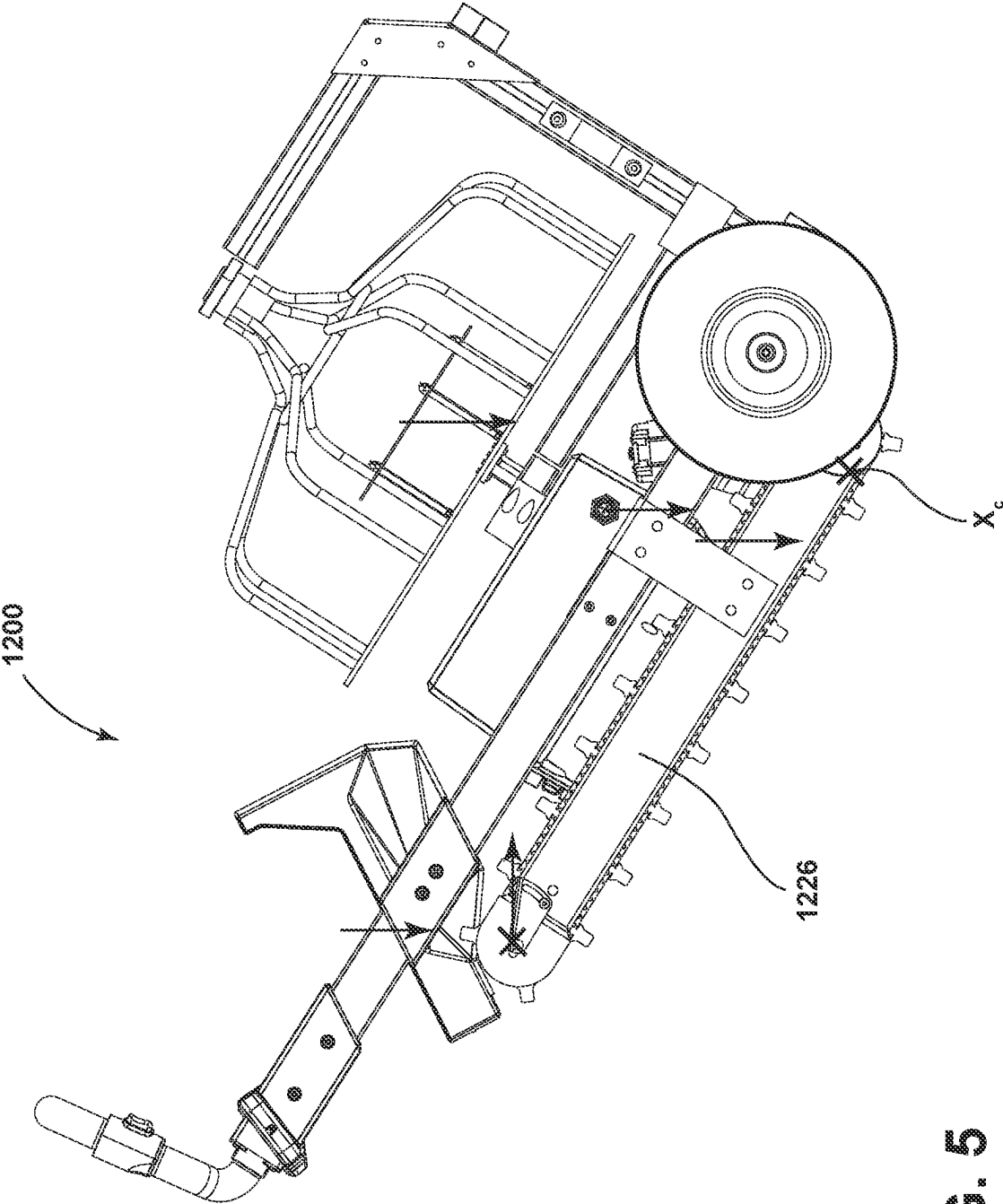


FIG. 5

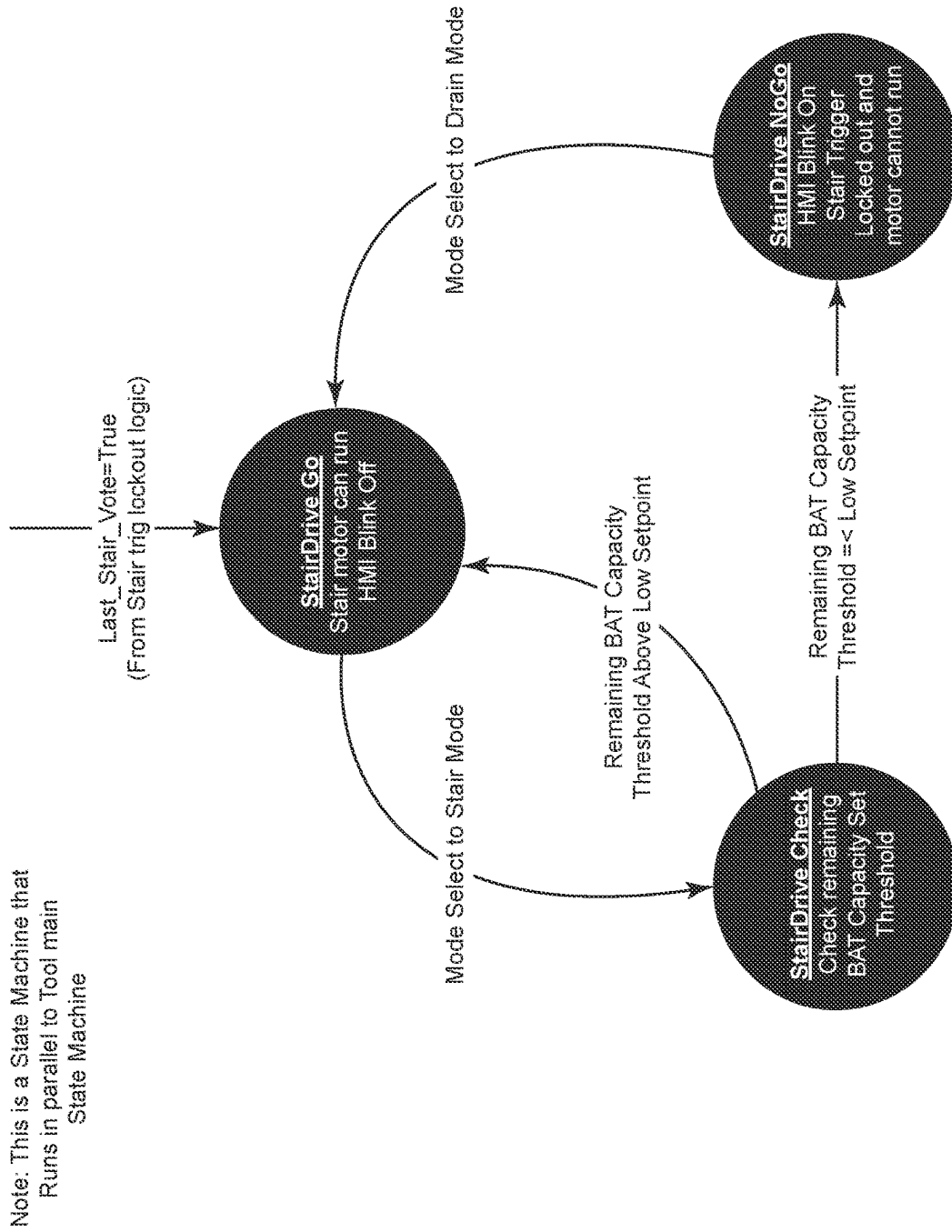


FIG. 6

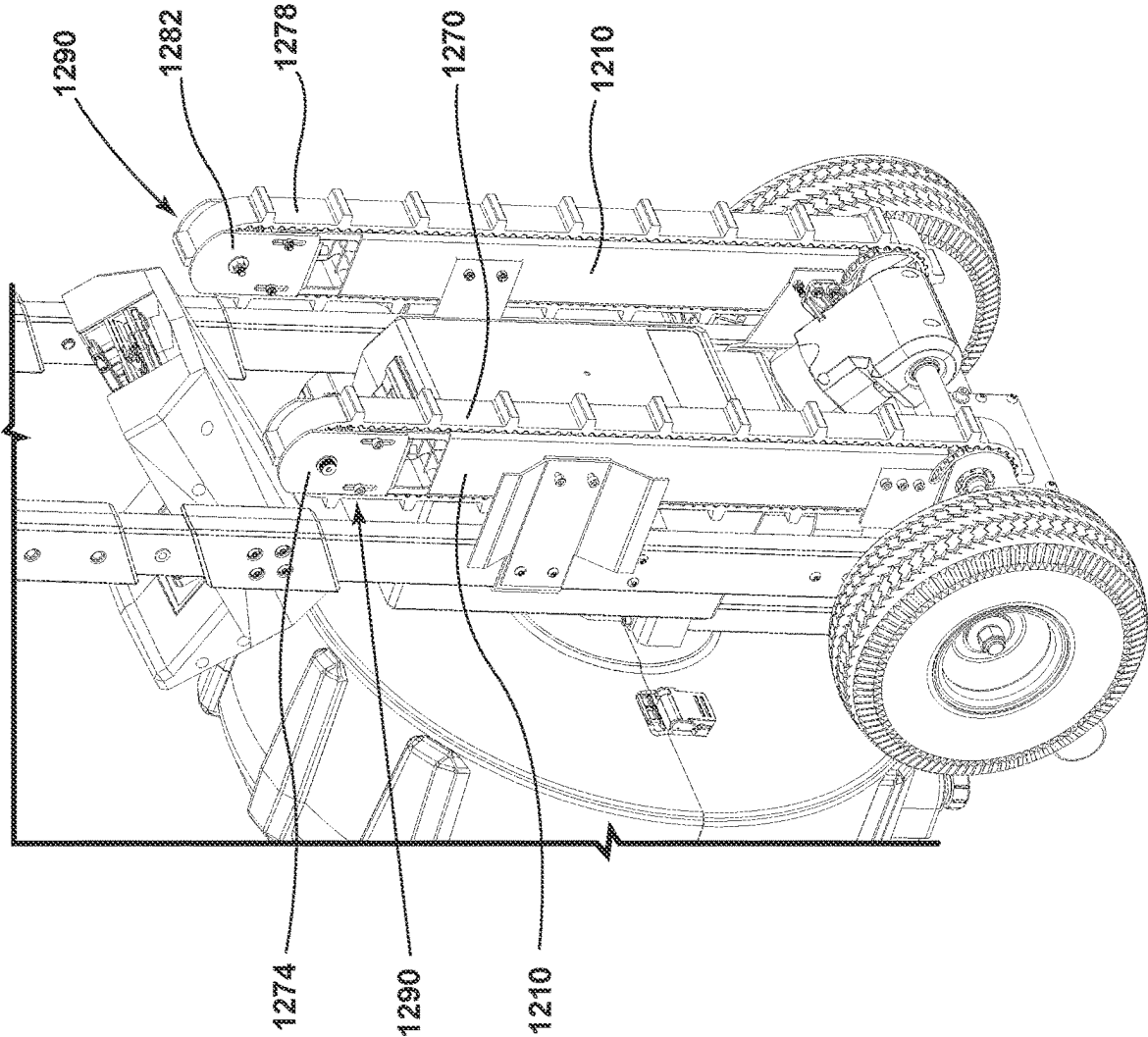
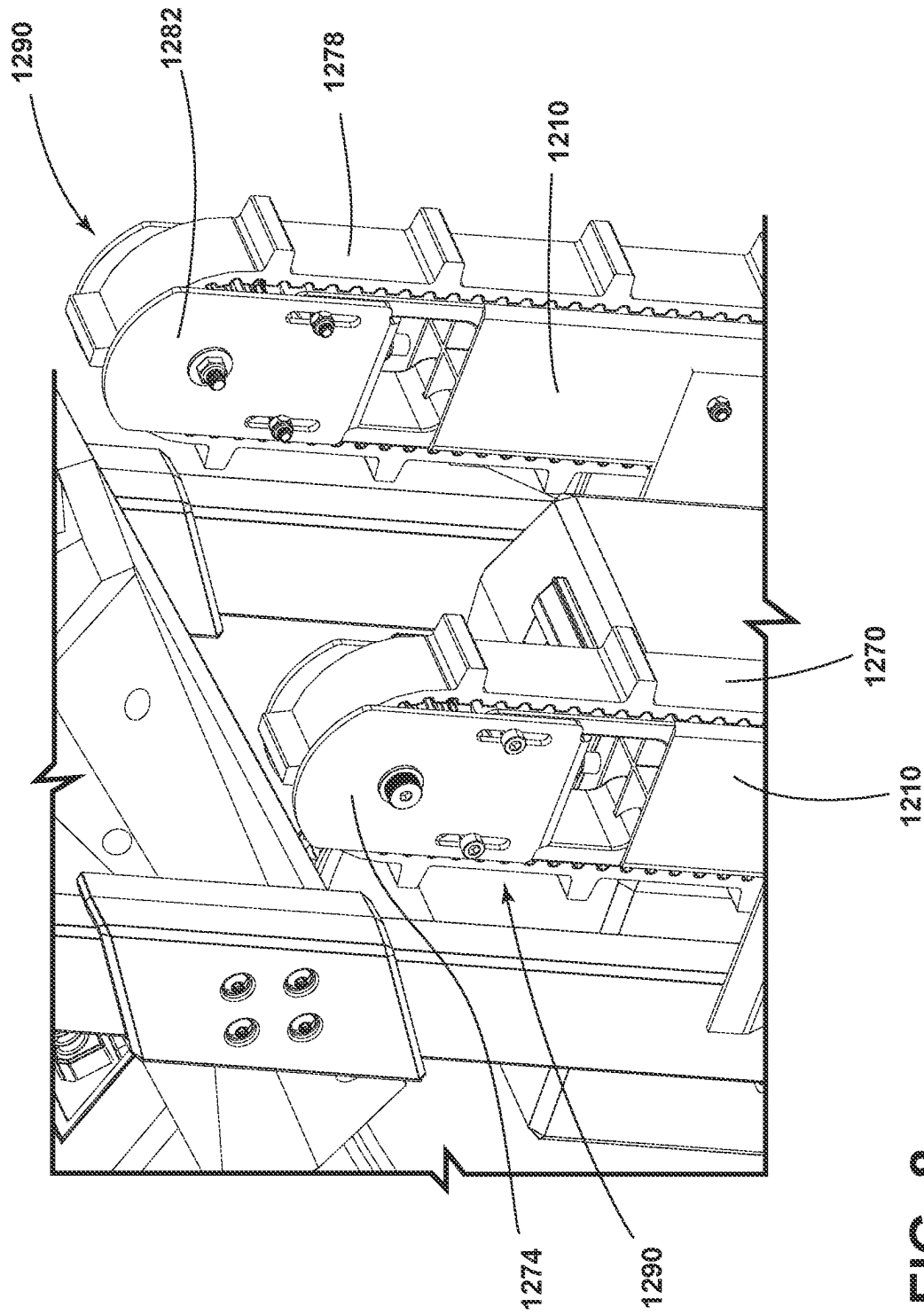


FIG. 7



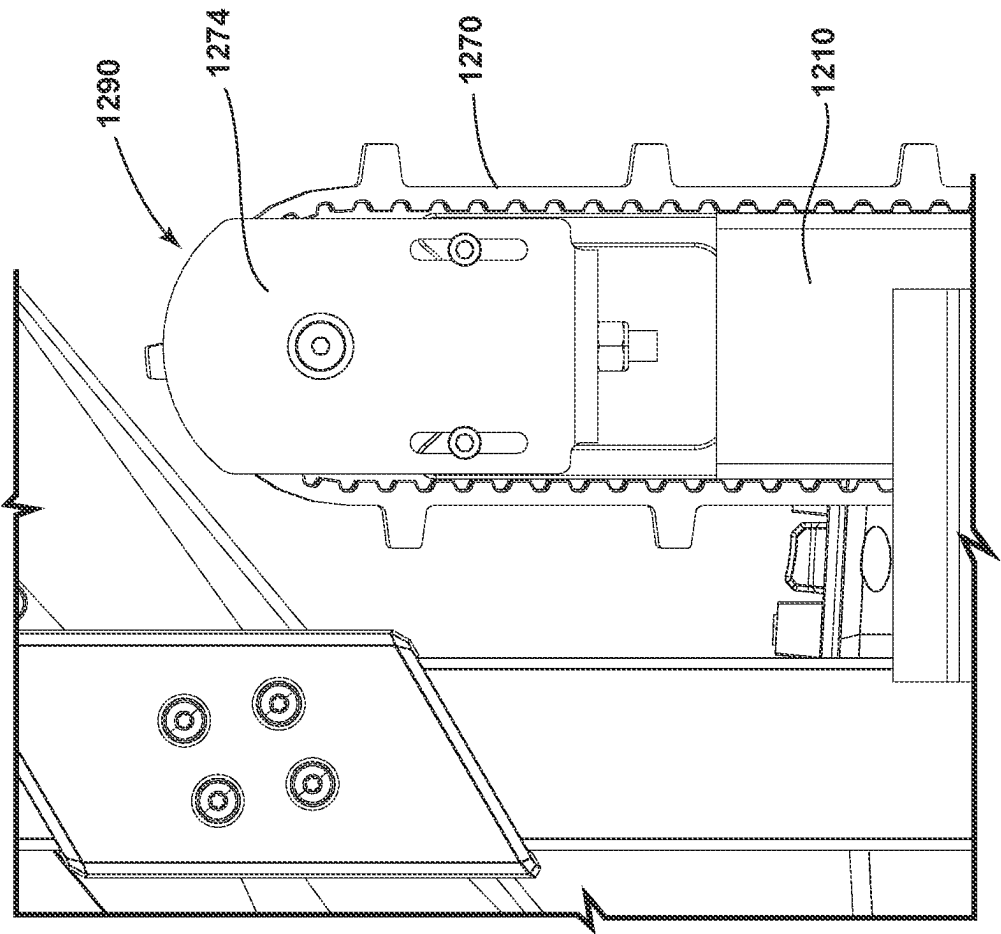


FIG. 9

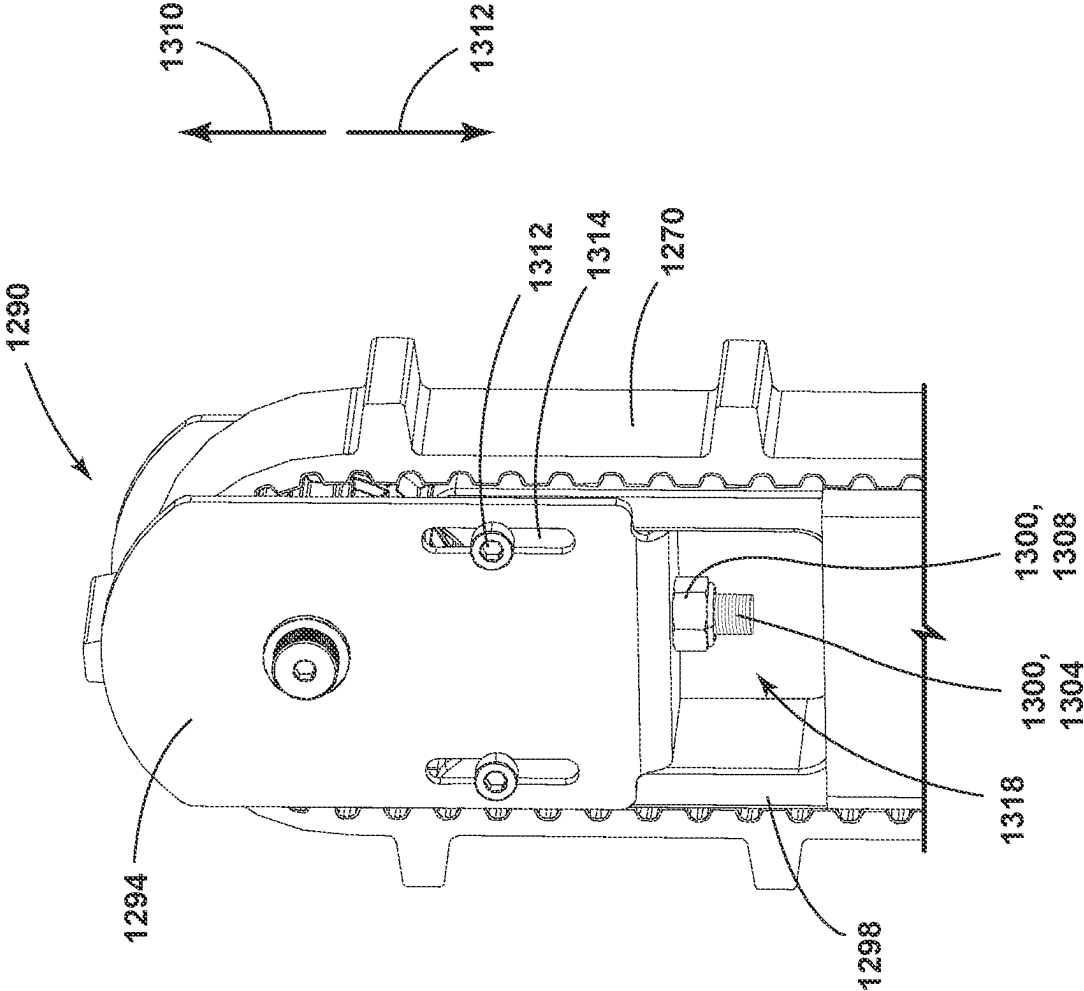


FIG. 10

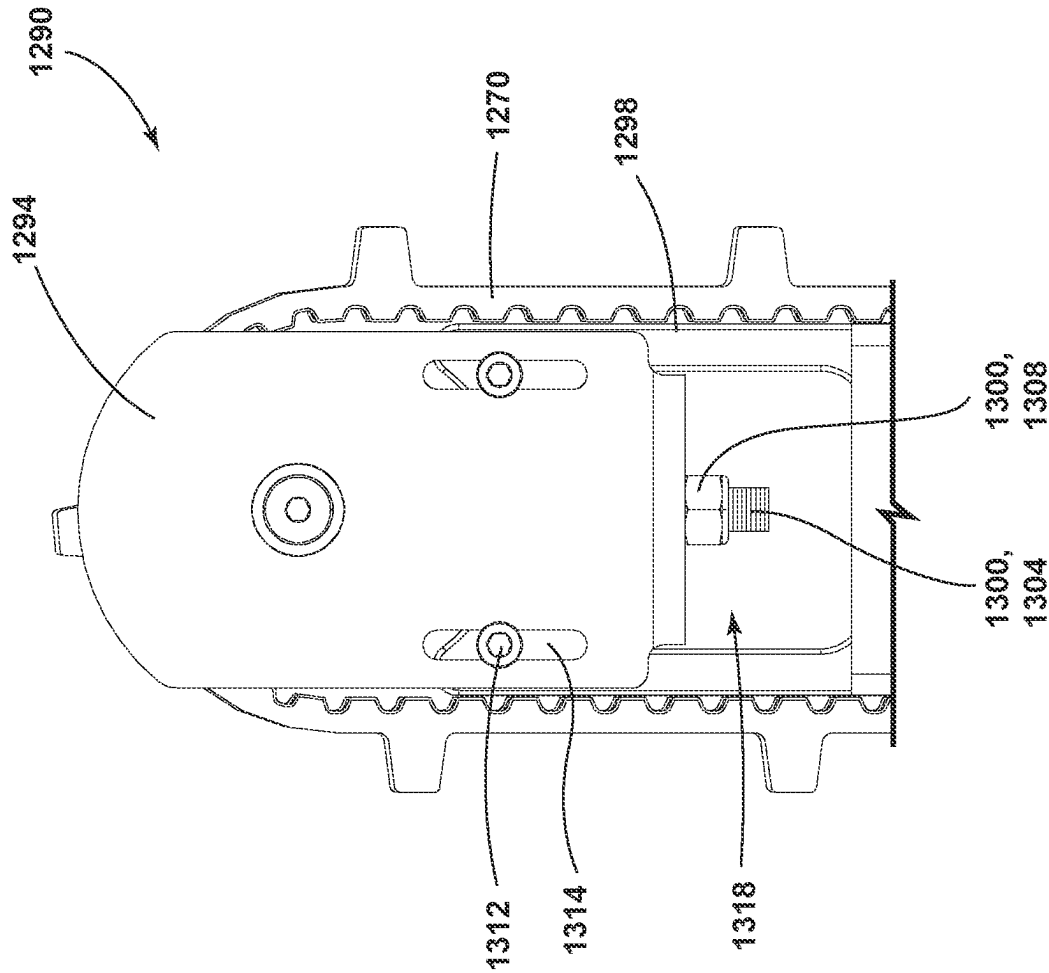


FIG. 11

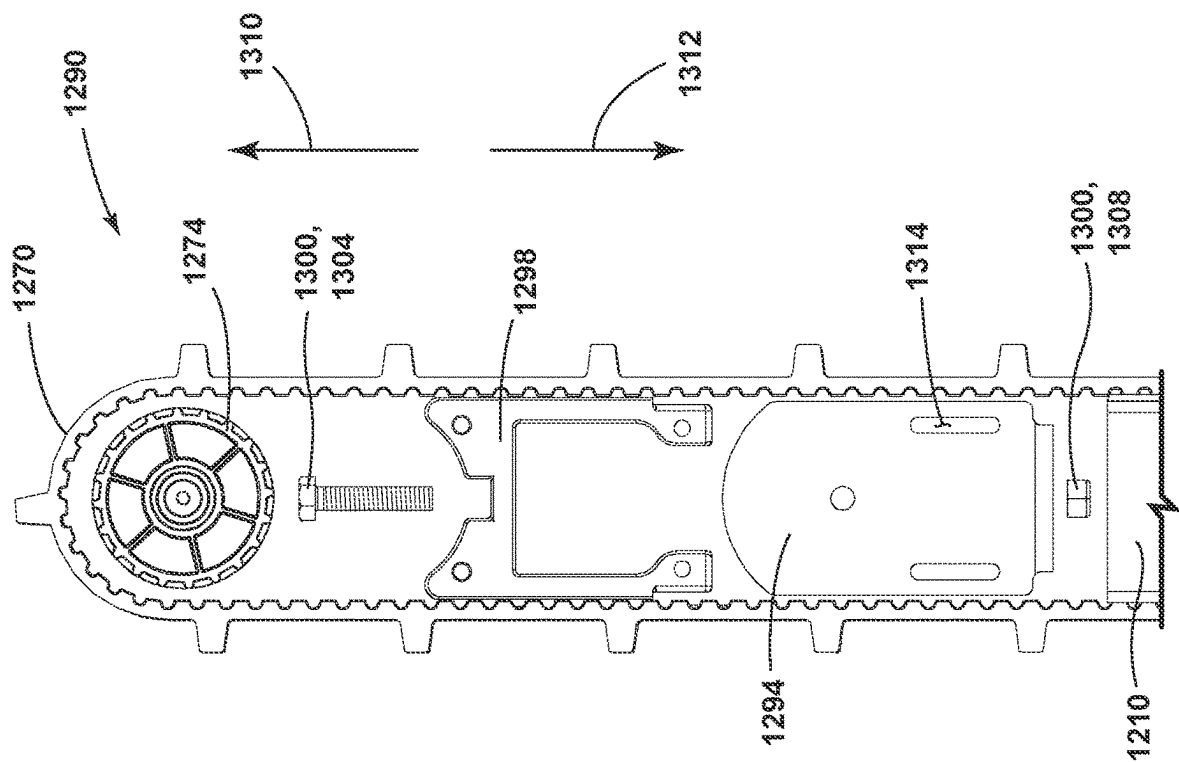
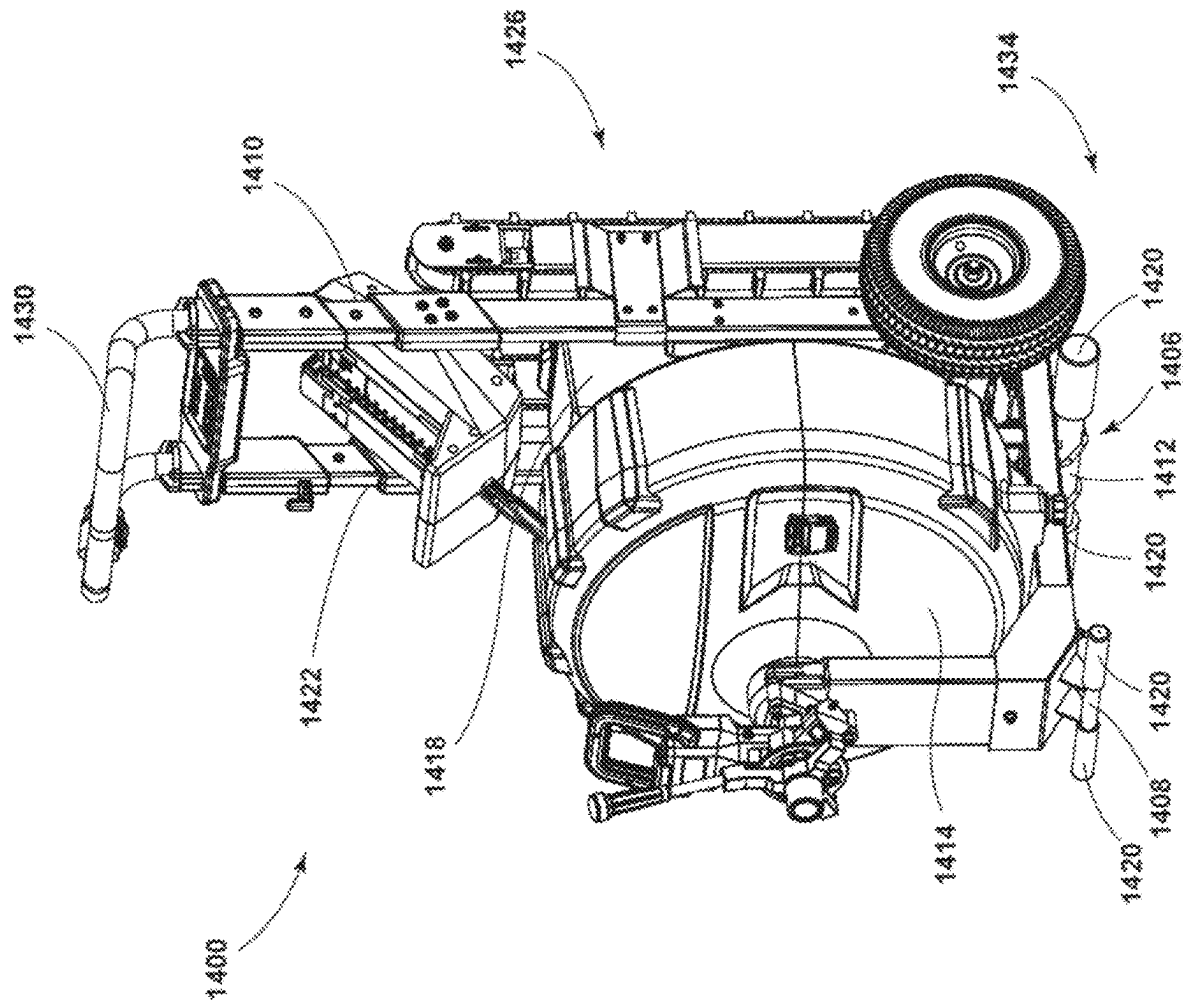


FIG. 12



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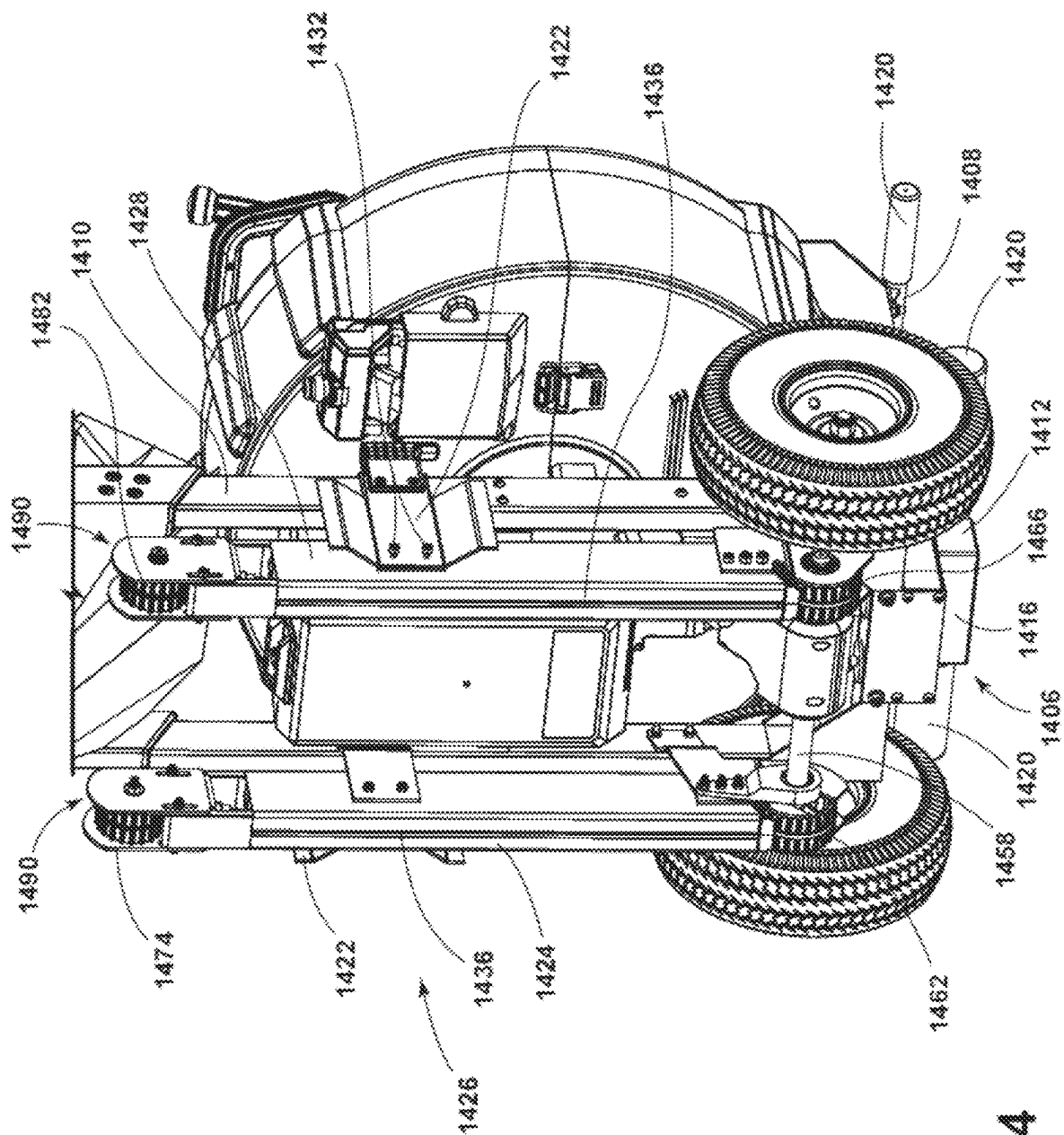
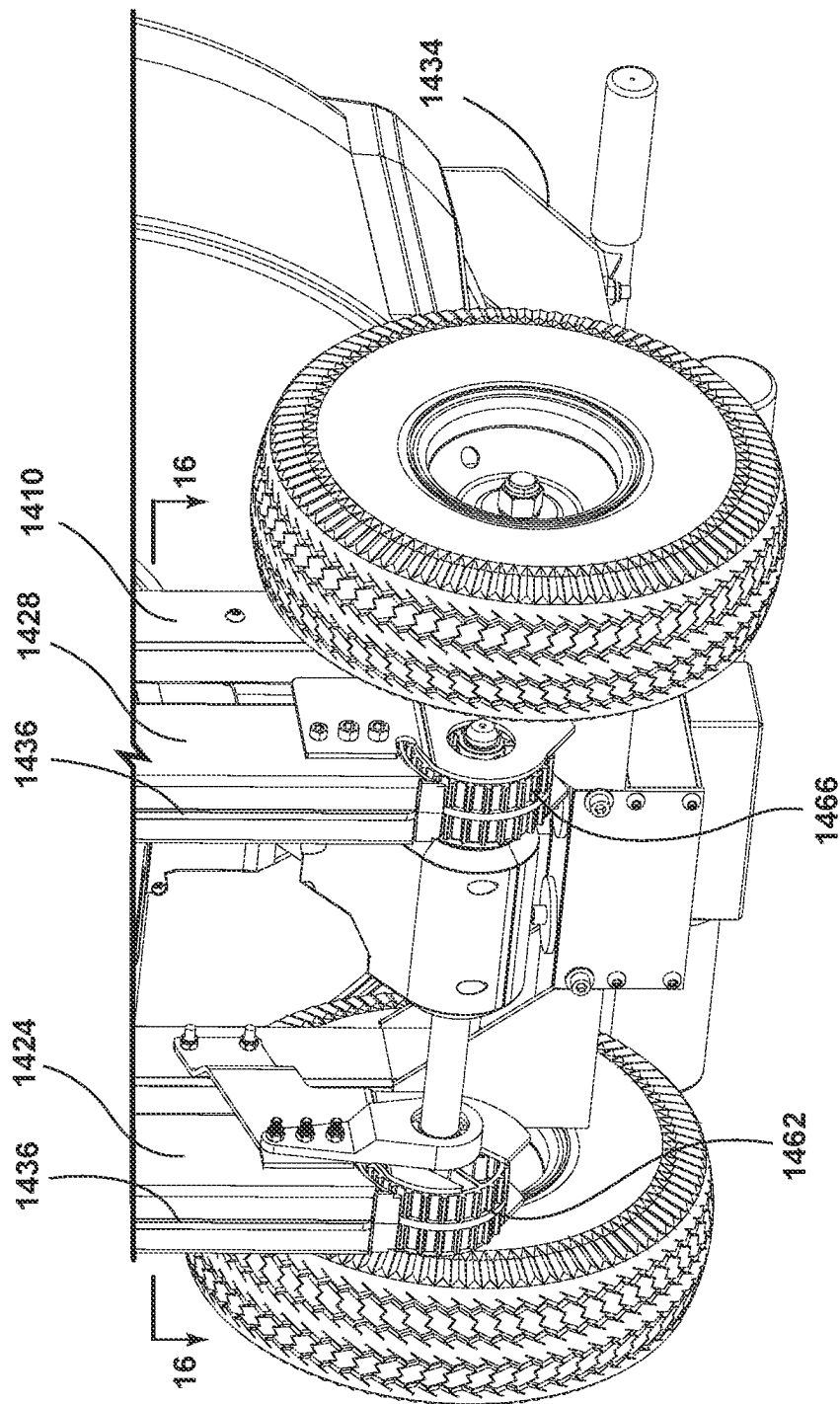


FIG. 14



57

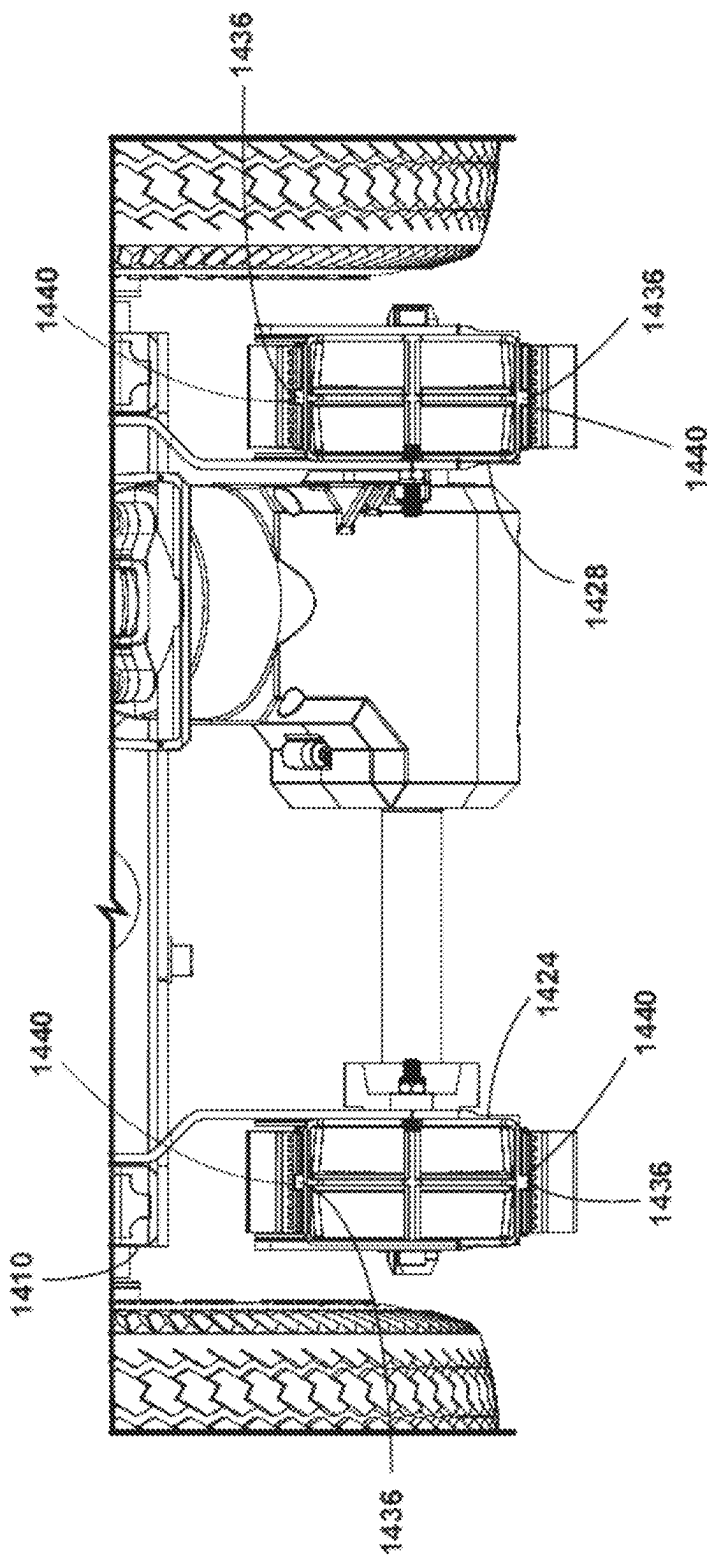


FIG. 16

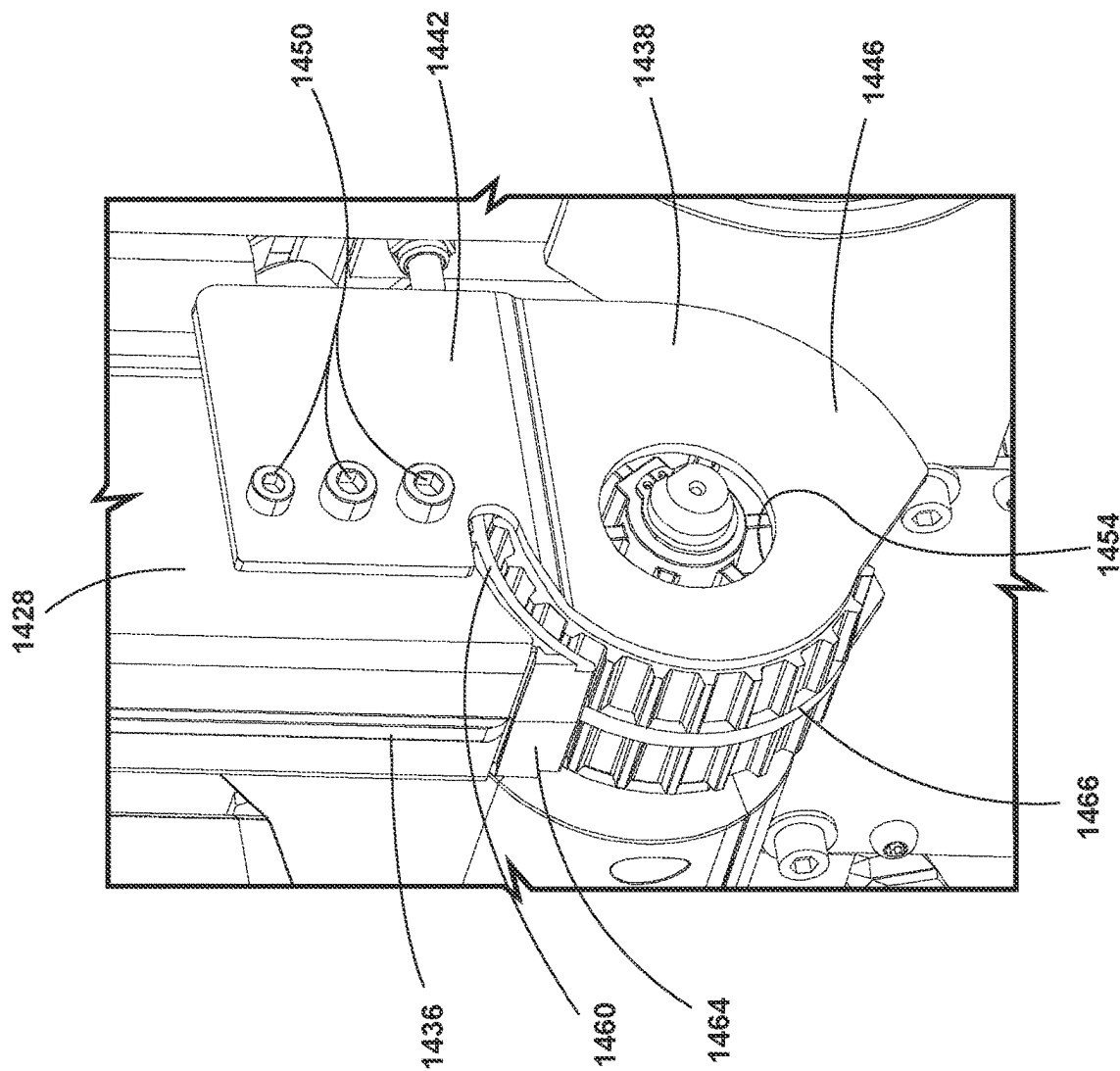


FIG. 17

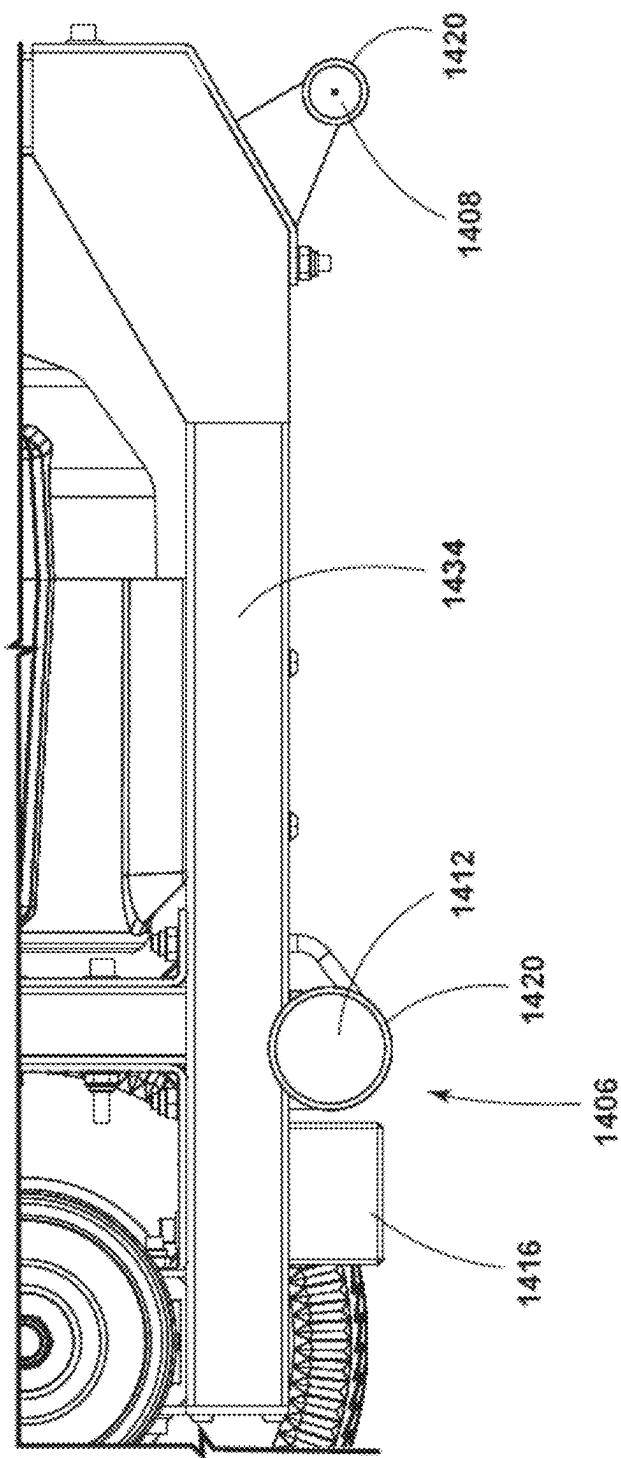


FIG. 18

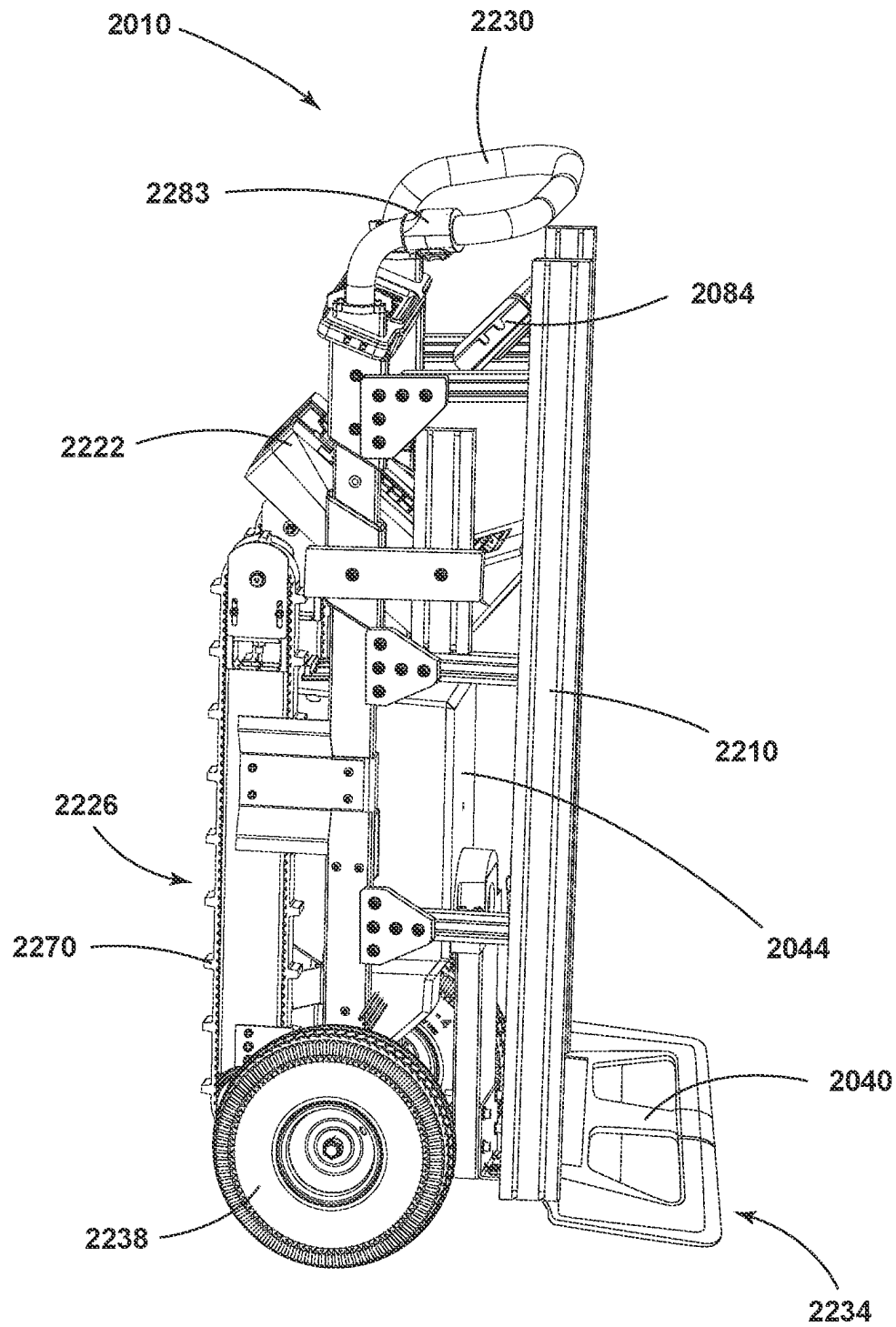


FIG. 19

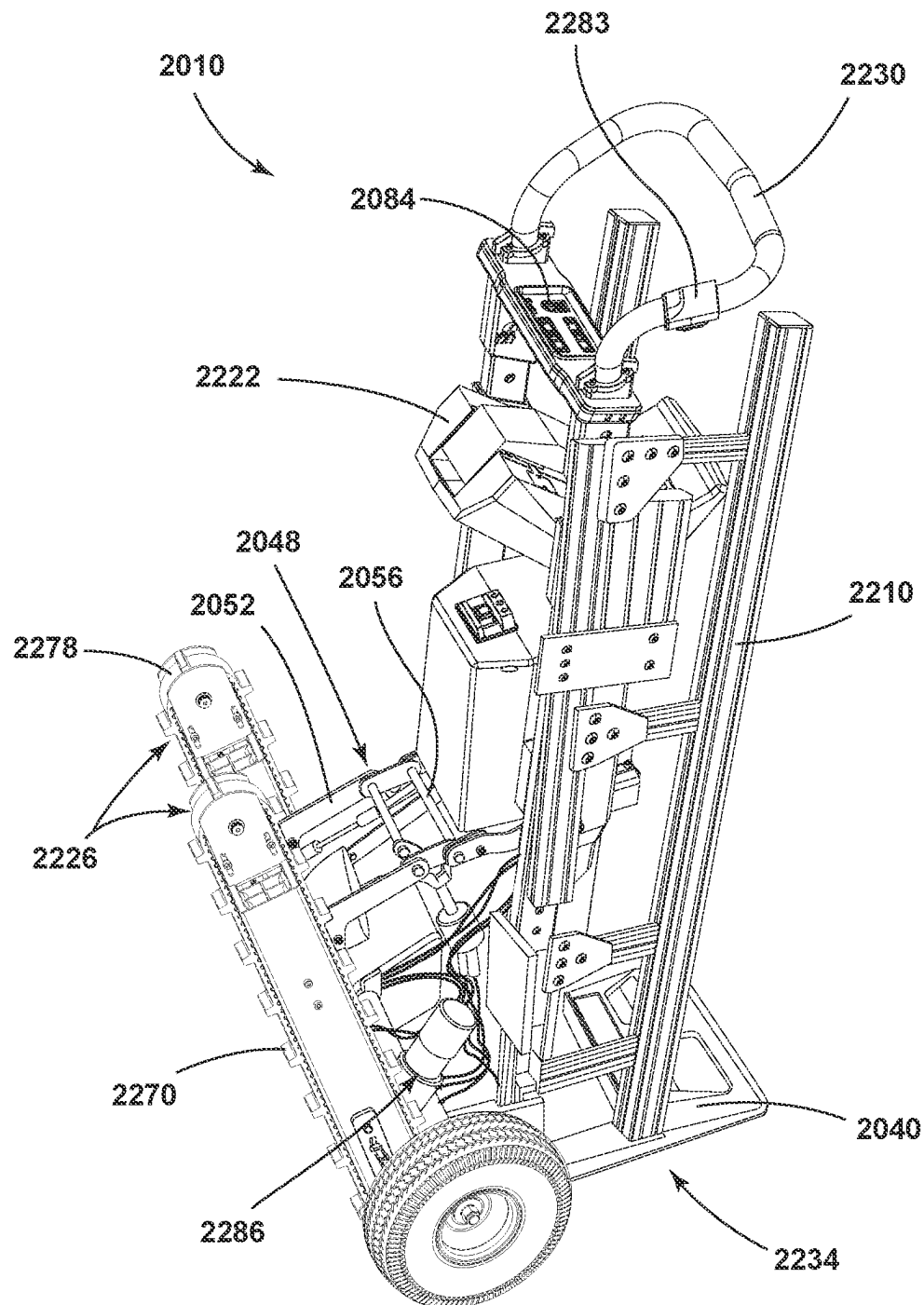


FIG. 20

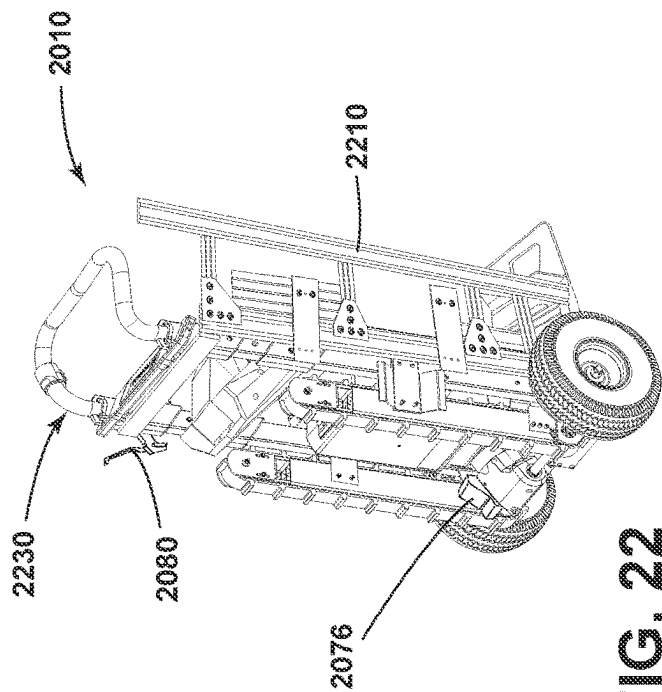


FIG. 22

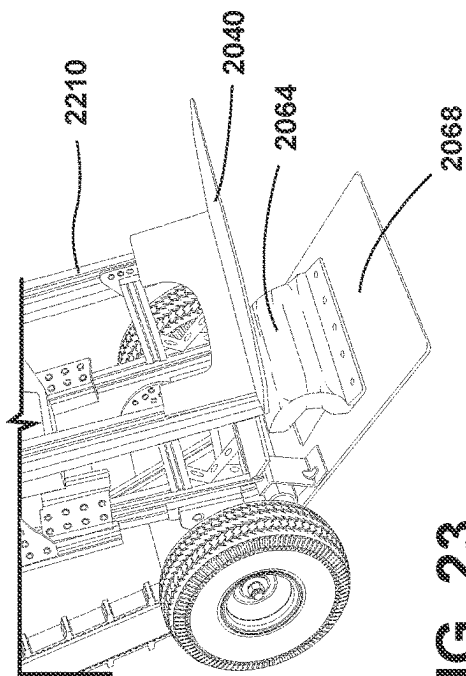


FIG. 23

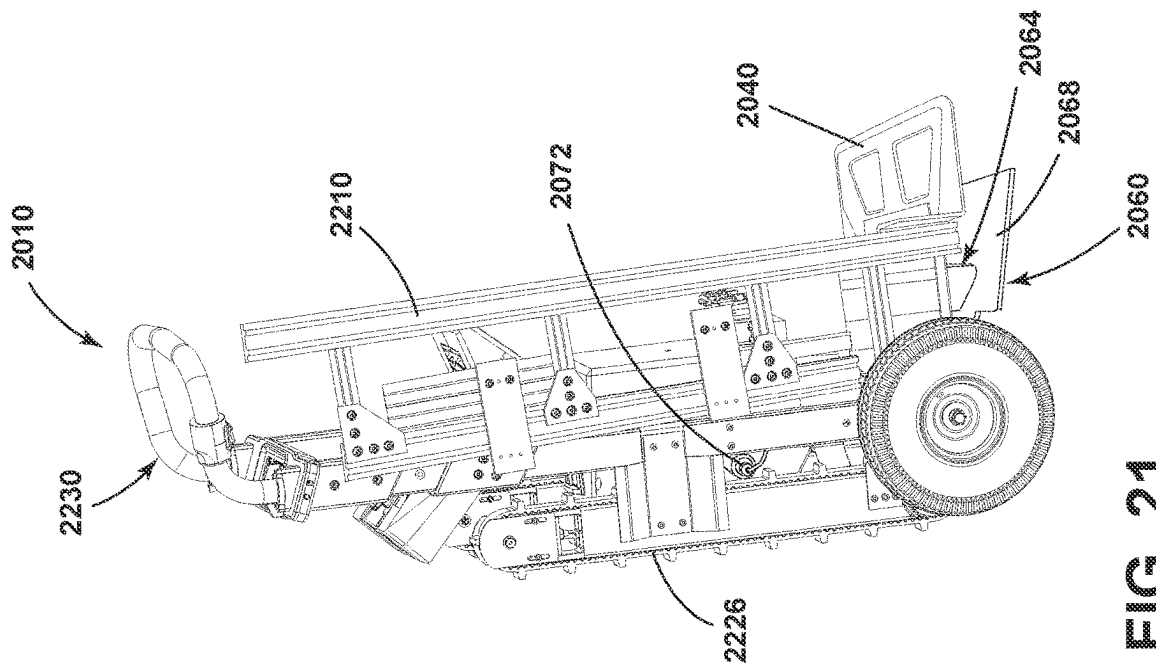


FIG. 21

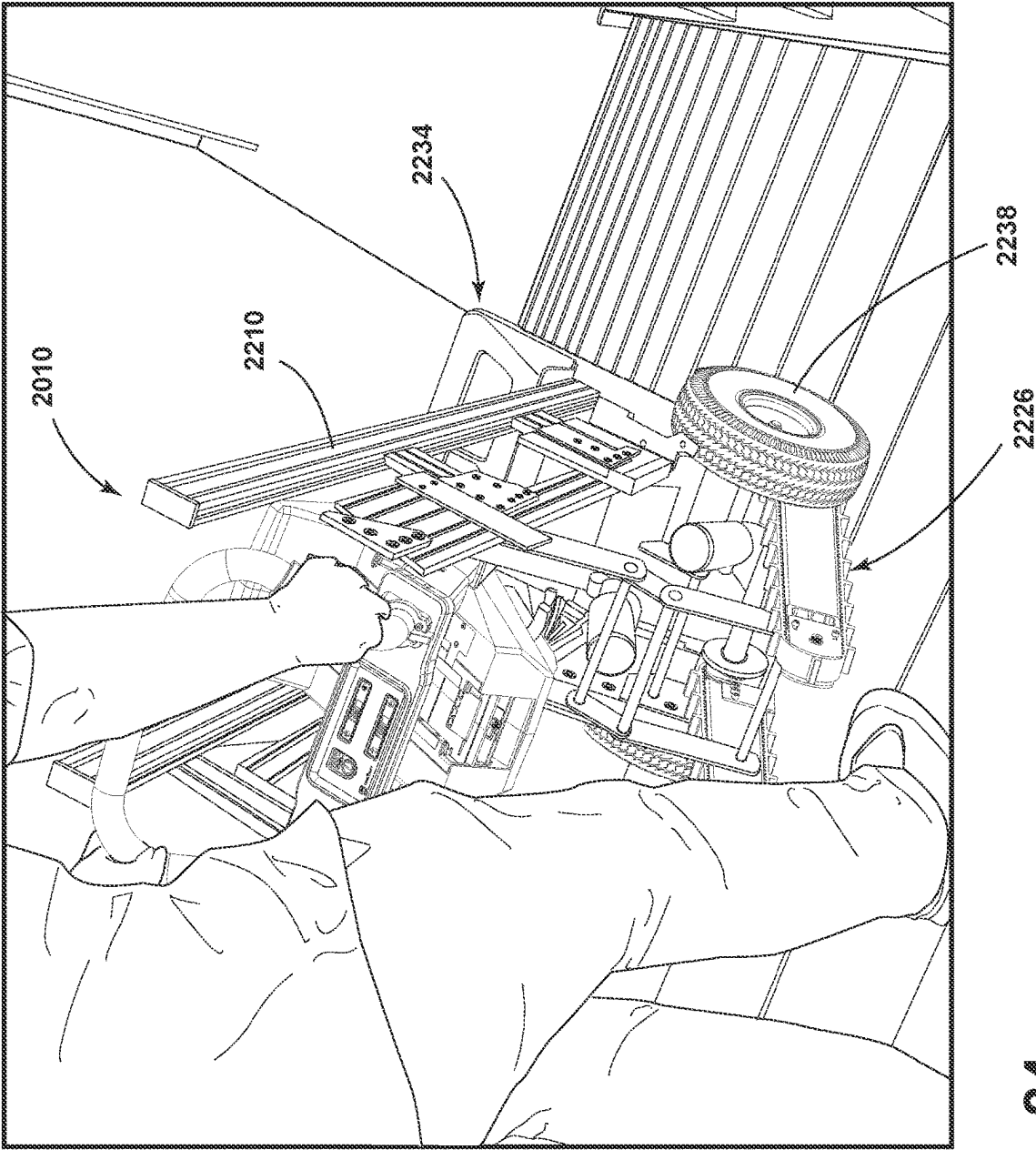


FIG. 24

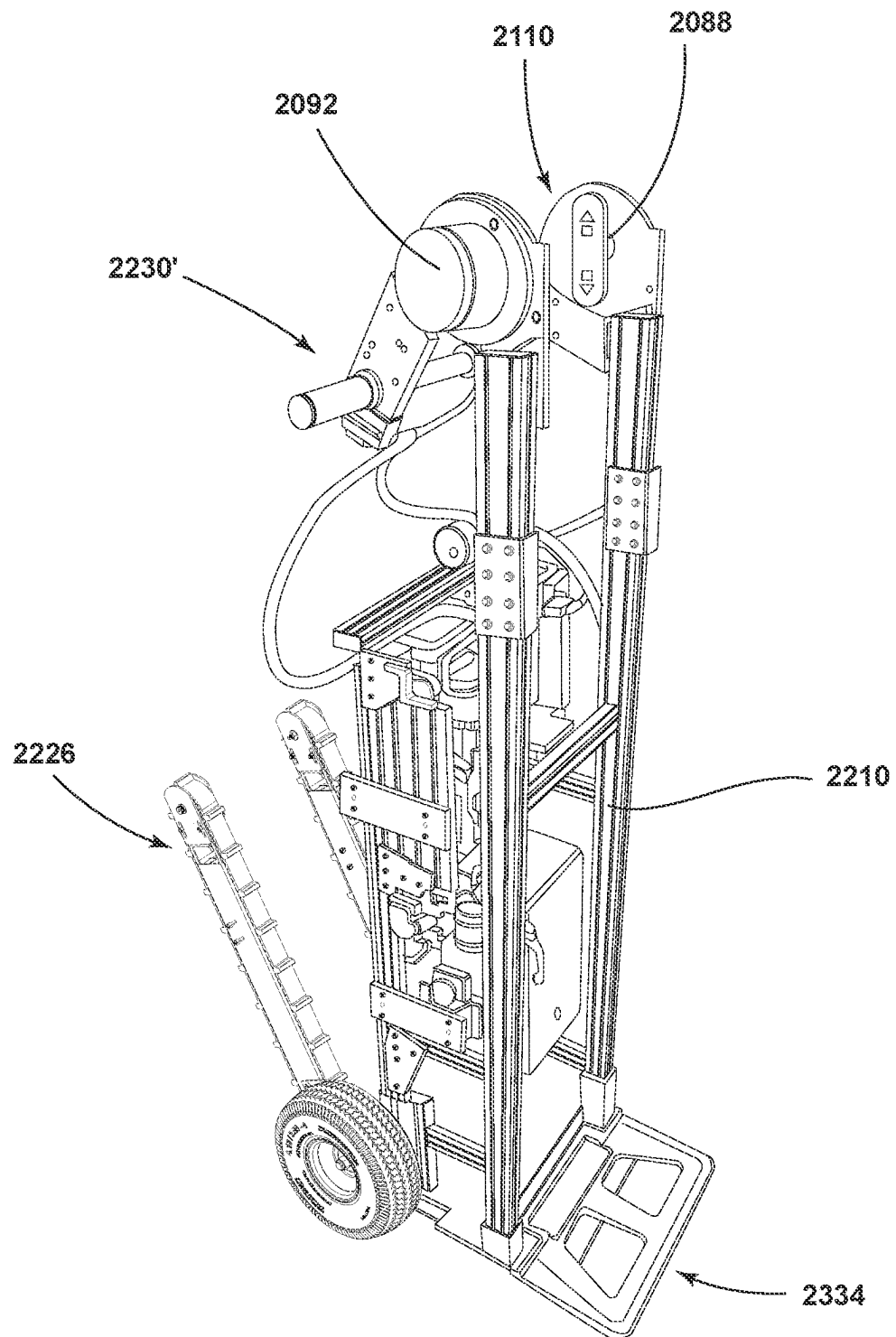


FIG. 25

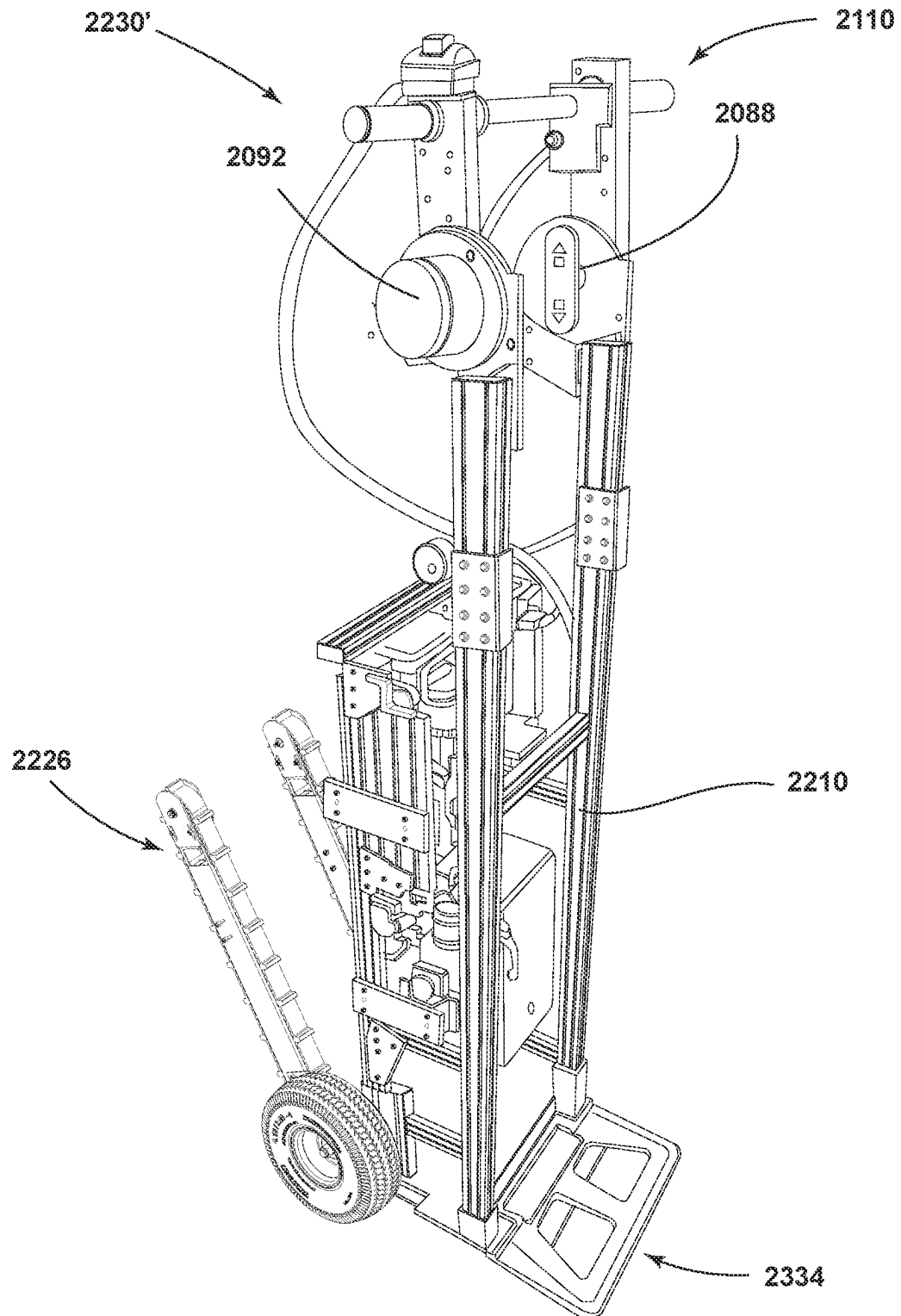


FIG. 26

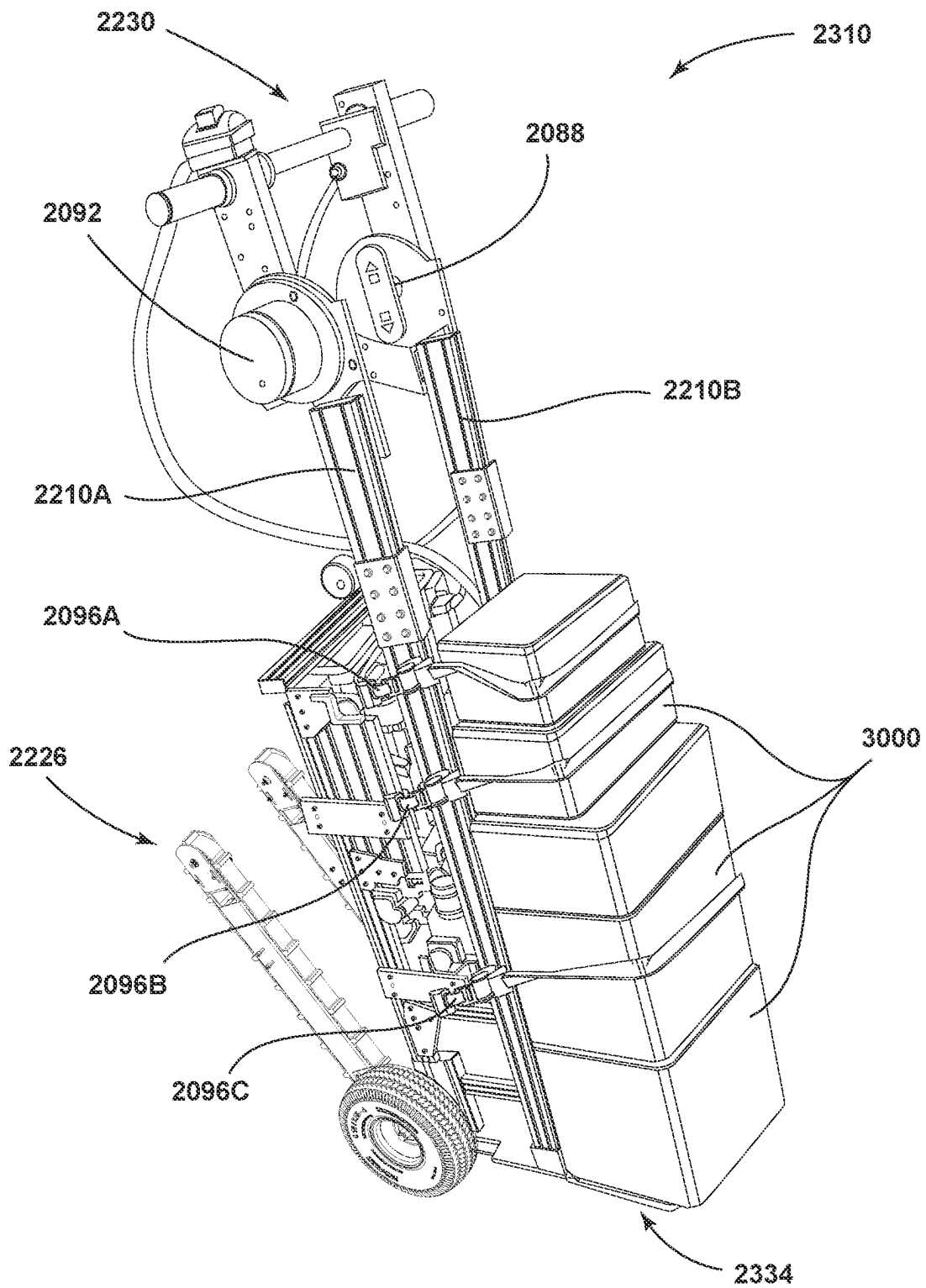
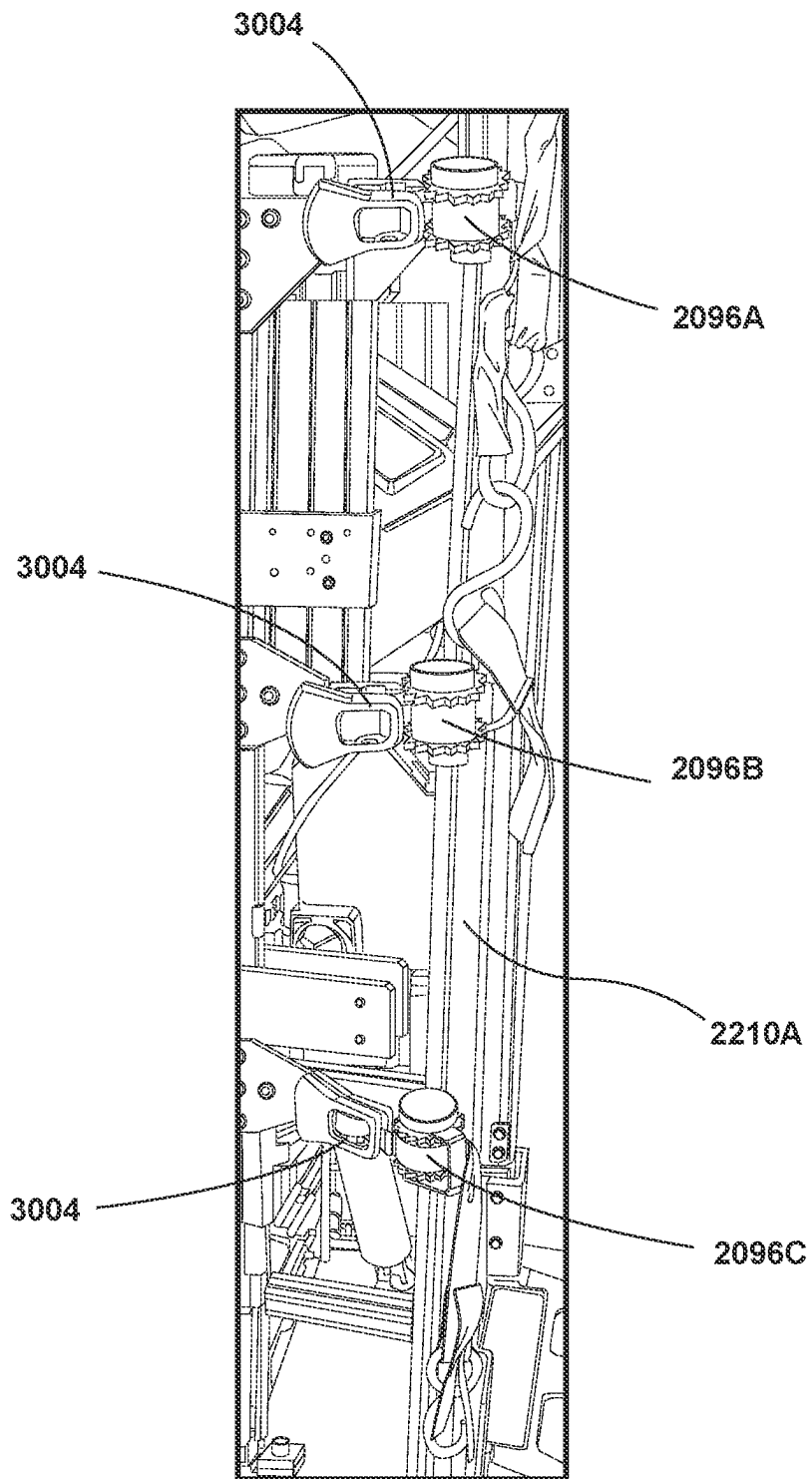


FIG. 27

**FIG. 28**

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TRANSPORTABLE MACHINE INCLUDING A TRACK SYSTEM

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a national phase filing under 35 U.S.C. 371 of International Application No. PCT/US2020/036914 filed Jun. 10, 2020, which claims priority to U.S. Provisional Patent Application No. 62/859,330 filed Jun. 10, 2019, U.S. Provisional Patent Application No. 62/924,533 filed Oct. 22, 2019, and U.S. Provisional Patent Application No. 62/987,383 filed Mar. 10, 2020, the entire contents of which are incorporated by reference herein.

FIELD OF INVENTION

The present disclosure relates to a dolly or hand truck for transporting a load. Furthermore, the present disclosure relates to sewer cleaning machines used to remove or dislodge debris caught in a drain or pipe.

BACKGROUND

Sewer cleaning machines are used to clean clogs and debris out of drains, sewers, and the like. Smaller handheld drain cleaners may be used to clean household drains from sinks or shower drains. However, larger and heavier cleaning machines are often used to clean sewers and industrial drains. A sewer cleaning machine may have as much as 200-300 feet of cable and a weight of 200-300 lbs. Accordingly, some sewer cleaning machines may be cumbersome to transport.

Hand trucks are used to transport a load from a first location to a second location. Hand trucks may be able to transport relatively large loads. Accordingly, some hand trucks may be helpful in transporting large machines or tools, such as a sewer cleaning machine. However, hand trucks also have limitations. For example, hand trucks are primarily helpful when transporting a load on a relatively flat or a smooth surface, or when the load does not get overly heavy. However, a hand truck may be cumbersome to transport when the load becomes too large, or when the hand truck must be used to transport a load up a set of stairs.

SUMMARY

In one embodiment, the present disclosure provides a transportable machine including an elongated frame having a top end and a bottom end, a handle disposed on the top end of the frame, a base disposed on the bottom end of the frame, where the base and the frame together define a space configured to receive and support a load, and a wheel disposed on the bottom end of the frame. A track is coupled to the frame, where the track includes a roller, an endless belt configured to rotate about the roller, and a belt tensioning assembly configured to tension the endless belt. The belt tensioning assembly includes a bracket supporting the roller, a mount coupled to the frame, and an adjustment member configured to adjust the position of the bracket relative to the mount. A motor is coupled to the roller, where the motor is operable to drive the endless belt.

In another embodiment, the present disclosure provides a transportable machine including an elongated frame having a top end and a bottom end, a handle disposed on the top end of the frame, a base disposed on the bottom end of the frame, where the base and the frame together define a space

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configured to receive and support a load, and a wheel disposed on the bottom end of the frame. A track is coupled to the frame, where the track includes a roller and an endless belt configured to rotate about the roller. A motor is coupled to the roller, where the motor is operable to drive the endless belt, a power supply provides power to the motor, and a controller is coupled to the power supply and the motor. The controller is configured to monitor the power supply, determine when the power supply is insufficient to climb a set of stairs, and in response to determining that the power supply is insufficient to climb the set of stairs, send a signal to alert a user, to shutoff the motor, or both.

In another embodiment, the present disclosure provides a transportable machine including an elongated frame having a top end and a bottom end, a handle disposed on the top end of the frame, a base disposed on the bottom end of the frame, where the base and the frame together define a space configured to receive and support a load, a wheel disposed on the bottom end of the frame, a track coupled to the frame, where the track includes a roller and an endless belt configured to rotate about the roller, and a motor coupled to the roller. The motor is operable to drive the endless belt in a first direction at a first speed and in a second direction at a second speed, where the second speed is slower than the first speed.

In another embodiment, the present disclosure provides a transportable machine including an elongated frame having a top end and a bottom end, a handle disposed on the top end of the frame, a base disposed on the bottom end of the frame, where the base and the frame together define a space configured to receive and support a load, a wheel disposed on the bottom end of the frame, and a track coupled to the frame. The track includes a roller and an endless belt configured to rotate about the roller, where the endless belt has a recess extending along an inside surface, and the recess receives a rib on the frame to align the endless belt relative to the frame. A motor is coupled to the roller, where the motor is operable to drive the endless belt.

In another embodiment, the present disclosure provides a transportable machine including an elongated frame having a top end and a bottom end, a handle disposed on the top end of the frame, a base disposed on the bottom end of the frame, where the base and the frame together define a space configured to receive and support a load, and a wheel disposed on the bottom end of the frame. A track is coupled to the frame, where the track includes a roller and an endless belt configured to rotate about the roller. A motor is coupled to the roller, where the motor is operable to drive the endless belt. A sewer cleaning machine is supported by the base, where the sewer cleaning machine includes a drum and a cable positioned at least partially within the drum. A support system is configured to support the transportable machine on a surface and inhibits the transportable machine from tipping, where the support system includes one or more supports positioned below the base.

In another embodiment, the present disclosure provides a transportable machine including an elongated frame having a top end and a bottom end, a handle disposed on the top end of the frame, a base disposed on the bottom end of the frame, where the base and the frame together define a space configured to receive and support a load, and a wheel disposed on the bottom end of the frame. A track is coupled to the frame, where the track includes a roller and an endless belt configured to rotate about the roller. The track is movable relative to the elongated frame between a stored position and an extended position, where the track is positioned farther away from the frame when in the extended

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position than in the stored position. A motor is coupled to the roller, where the motor is operable to drive the endless belt.

In another embodiment, the present disclosure provides a transportable machine including an elongated frame having a top end and a bottom end, a handle disposed on the top end of the frame, a base disposed on the bottom end of the frame, where the base and the frame together define a space configured to receive and support a load, and a wheel disposed on the bottom end of the frame. A track is coupled to the frame, where the track includes a roller and an endless belt configured to rotate about the roller. A motor is coupled to the roller, where the motor is operable to drive the endless belt. A lifting system is configured to tilt the base relative to a support surface.

The above aspects may be implemented in different combinations and not necessarily in the ordered presented. Other aspects of the disclosure will become apparent by consideration of the detailed description and accompanying drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a front perspective view of a sewer cleaning machine.

FIG. 2 is a side view of the transportable machine of FIG. 1.

FIG. 3 is a rear perspective view of the transportable machine of FIG. 1 with a motor housing removed.

FIG. 4 is a cross-sectional view of the transportable machine of FIG. 1.

FIG. 5 is a side view of a frame of the transportable machine of FIG. 1.

FIG. 6 is a flow chart of a stair-drive check.

FIG. 7 is a perspective view of a transportable machine with a belt tensioning mechanism according to one embodiment.

FIG. 8 is a perspective view of the belt tensioning mechanism of FIG. 7.

FIG. 9 is a side view of the belt tensioning mechanism of FIG. 7.

FIG. 10 is an enlarged perspective view of the belt tensioning mechanism.

FIG. 11 is an enlarged side view of the belt tensioning mechanism of FIG. 10.

FIG. 12 is an exploded view of the belt tensioning mechanism of FIG. 10.

FIG. 13 is a front perspective view of a transportable machine according to another embodiment of the disclosure.

FIG. 14 is a rear perspective view of the transportable machine of FIG. 13 with belts removed.

FIG. 15 is an enlarged view of the transportable machine of FIG. 14.

FIG. 16 is a cross-sectional view of the transportable machine of FIG. 15 taken along lines 16-16.

FIG. 17 is a perspective view of a portion of the transportable machine of FIG. 13.

FIG. 18 is a side view of a portion of the transportable machine of FIG. 13.

FIG. 19 is a perspective view of a transportable machine, illustrating a track of the transportable machine in a stored position.

FIG. 20 is another perspective view of the transportable machine of FIG. 19, illustrating the track of the transportable machine in an extended position.

FIG. 21 is another perspective view of the transportable machine of FIG. 19.

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FIG. 22 is a rear perspective view of the transportable machine of FIG. 19, illustrating an air bag member.

FIG. 23 is a partial perspective view of a base of the transportable machine of FIG. 21.

FIG. 24 is a perspective view of the transportable machine of FIG. 19, illustrating movement of the transportable machine by a user relative to stairs.

FIG. 25 is a perspective view of another transportable machine embodying the disclosure, illustrating a handle of the transportable machine in a first position.

FIG. 26 is another perspective view of the transportable machine of FIG. 25, illustrating the handle in a second position.

FIG. 27 is a perspective view of yet another transportable machine embodying the disclosure, illustrating a securing mechanism.

FIG. 28 is a partial side view of the transportable machine of FIG. 27.

DETAILED DESCRIPTION

Before any embodiments of the disclosure are explained in detail, it is to be understood that the disclosure is not limited in its application to the details of construction and the arrangement of components set forth in the following description or illustrated in the following drawings. The disclosure is capable of other embodiments and of being practiced or of being carried out in various ways.

FIGS. 1-4 illustrate a transportable machine according to one embodiment. The transportable machine illustrated in FIGS. 1-4 is a sewer cleaning machine 1200. As will be understood, the features shown and described herein with respect to the other embodiments of a transportable machine may be used in combination with the features shown and described with respect to the sewer cleaning machine 1200.

The sewer cleaning machine 1200 includes a frame 1210, a handle 1230, a base 1234, and a track 1226. The frame 1210 defines an elongated body extending between the handle 1230 and the base 1234. Specifically, the handle 1230 is positioned on a top end of the frame 1210 to enable a user to grip and control movement of the sewer cleaning machine 1200. The base 1234 is positioned on a bottom end of the frame 1210 and is configured to support a load, such as a machine or other load, that may be desirable to be able to transport. For example, the base 1234 may be configured to support many of the components of the sewer cleaning machine 1200. In the embodiment illustrated in FIGS. 1-4, the base 1234 supports a drum housing 1214, a motor housing 1218, and a power supply 1222. The track 1226 is supported by the frame 1210 to help facilitate transport of the sewer cleaning machine 1200. In the illustrated embodiment, the track 1226 is positioned on a rear side of the frame 1210 between the handle 1230 and the base 1234. Additionally, the illustrated sewer cleaning machine 1200 includes wheels 1238 rotatably coupled to the frame 1210 to also assist with transport of the sewer cleaning machine 1200.

In the illustrated embodiment, a drum 1242 is rotatably supported within the drum housing 1214 and includes a cable (not shown in FIGS. 1-4) that is extendable out of an opening 1246 on the drum 1242. In some embodiments, the drum 1242 is a cage-style drum that when the drum housing 1214 is opened allows easy access to the cable so a user can inspect the cable. The cable is extendable out of the drum with a cable feed device 1250. A first motor 1254 is supported within the motor housing 1218 and is coupled to the drum 1242. The first motor 1254 is operable to rotate the drum 1242. Rotation of the drum 1242 creates friction

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between an inner surface of the drum 1242 and the cable, which causes the cable to spin to facilitate clearing debris from a drain pipe or another conduit.

The track 1226 is configured to engage a surface, such as stairs or a ramp to help a user maneuver the sewer cleaning machine 1200. In the illustrated embodiment, the track 1226 is positioned on the opposite side of the frame 1210 as the drum 1242. For example, the drum 1242 is positioned on a front side of the frame 1210, and the track 1226 is positioned on a back side of the frame 1210. In other embodiments the track 1226 can be positioned on either the left or right side of the frame 1210.

The track 1226 includes a substantially horizontal drive shaft 1258 with a first drive roller 1262 at one end and a second drive roller 1266 at another end. A first endless belt 1270 extends around the first drive roller 1262 and a first idler roller 1274, and a second endless belt 1278 extends around the second drive roller 1266 and a second idler roller 1282. The endless belts 1270, 1278 extend substantially vertically along a length of the frame 1210. In some embodiments, the track 1226 may only include a first endless belt 1270 and respective rollers 1262, 1274 rather than including first and second endless belts 1270, 1278. In some embodiments, the endless belts 1270, 1278 include traction elements that assists in gripping a surface, ledge, or other object. For example, in the illustrated embodiment, the endless belts 1270, 1278 include castellations 1276, or projections, that help grip various surfaces. In further embodiments, the endless belts 1270, 1278 include replaceable cleats that can be replaced when worn down instead of replacing the entire track 1226. In other embodiments, the track 1226 can articulate or expand and retract to better climb stairs.

As shown in FIG. 3, a second motor 1286 is supported by the frame 1210 and is coupled to the drive shaft 1258 of the track 1226. The second motor 1286 is operable to rotate the drive shaft 1258 and thus the drive rollers 1262, 1266 to facilitate rotation of the endless belts 1270, 1278. In other embodiments, the track 1226 is passively driven rather than motor driven. For example, the endless belts 1270, 1278 of the track 1226 may be rotated by engagement with a surface, such as stairs or a ramp, as a user pulls the sewer cleaning machine 1200. Regardless of whether the track 1226 is motor driven or passively driven, the track 1226 is decoupled from the drum 1242 in that the track 1226 is independently rotated without affecting the rotation of the drum 1242.

In the illustrated embodiment, both the first and second motors 1254, 1286 are powered by the power supply 1222 that is supported on the frame 1210. The first and second motors 1254, 1286 are, for example, brushless motors. In additional embodiments, the first and second motors 1254, 1286 are variable two speed motors. In the illustrated embodiment, the power supply 1222 includes a battery receptacle that receives a battery pack to provide D/C power to the sewer cleaning machine 1200. For example, the battery receptacle may removably receive a rechargeable power tool battery pack. In further embodiments, the power supply 1222 may receive more than one battery pack to power the sewer cleaning machine 1200. In alternative embodiments, the power supply 1222 may be coupled to a power outlet to provide A/C power to the sewer cleaning machine 1200.

The power supply 1222 is electrically coupled to a controller 1284 that may control operation of the first and second motors 1254, 1286. In some embodiments, the controller 1284 ensures that when one motor is operating the

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other motor is locked out and cannot be run. As previously mentioned, the track 1226 is decoupled from the drum 1242 such that rotation of one is independent of the other. In some embodiments, the controller 1284 actively decouples the track 1226 from the drum 1242 so that they cannot operate at the same time. In other embodiments, the track 1226 is decoupled from the drum 1242 only by lack of mechanical connection to the drum 1242. Additionally, the sewer cleaning machine 1200 may include switches, buttons, a user interface, or other control features that allow a user to selectively control the sewer cleaning machine 1200.

Further, the power supply 1222 or the battery may include a battery fuel gauge to indicate to a user how much longer the battery will last. Additionally, the sewer cleaning machine 1200 may include battery detection that indicates to a user if the sewer cleaning machine 1200 has enough power to climb a standard set of stairs and, if not, lock out the tracks 1226 from being operated. This inhibits a user from initiating a stair climb and then running out of power in the middle of the stairs where it is difficult to manually maneuver the sewer cleaning machine 1200.

Referring to FIG. 6, a user can adjust the sewer cleaning machine 1200 from a drain cleaning mode, in which the drum 1242 is operable, to a stair climbing mode, in which the track 1226 is operable. When the user adjusts the sewer cleaning machine to the stair climbing mode, the control system of the sewer cleaning machine 1200 initiates a stair-drive check to determine whether the power supply 1222 has sufficient power to climb a set of stairs. When the remaining capacity of the power supply 1222 is sufficient to climb an entire set of stairs, the system will allow the second motor 1286 to power the track 1226 and enable the sewer cleaning machine 1200 to climb a set of stairs. Specifically, when the capacity of the power supply 1222 is above a predetermined threshold, the track 1226 is operable to climb the stairs. When the remaining capacity of the power supply 1222 is insufficient to climb an entire set of stairs, the system enters a lock out mode and prevents the second motor 1286 from powering the track 1226. Specifically, when the capacity of the power supply 1222 is below a predetermined threshold, the track 1226 is locked out. Alternatively, when the remaining power supply 1222 is insufficient to climb an entire set of stairs, the system provides an alert to the user indicating that the battery power is insufficient to client the entire set of stairs. In this embodiment, the user may have the option to continue to utilize the track 1226 in a driven mode until the power supply 1222 is completely depleted and then may climb the remaining stairs using the track 1226 in an idle mode.

The predetermined threshold can be determined based on a standard height of a set of stairs. In some embodiments, the predetermined threshold can be determined based on a variety of different sized stair sets, such as a half stair well or an extra long stair well. When the sewer cleaning machine 1200 enters the lock out mode, an indicator informs the user that there is insufficient battery to complete the stair climb. The indicator may include, for example, a light on the sewer cleaning machine 1200 and/or an audible alert. The user may then charge the power supply 1222 or swap battery packs so that the capacity exceeds the predetermined threshold. When the sewer cleaning machine 1200 is in the locked out mode, a user has the option to adjust the sewer cleaning machine back to a drain cleaning mode. Although the power supply may not be sufficient to initiate a stair climb, the drum 1242 may still be operated when the capacity of the power supply 1222 is below the predetermined threshold.

Referring back to FIGS. 2-4, the sewer cleaning machine 1200 is shown in a first operational position. In this position, the drum housing 1214 of the sewer cleaning machine 1200 is supported on a surface to facilitate the clearing of debris from a conduit. In operation, the power supply 1222 supplies power to the first motor 1254 to spin the drum 1242. The cable feed device 1250 draws cable from inside the drum 1242 so that a user may extend the cable into a drain. Rotation of the drum 1242 causes the cable to spin, assisting in the removal of debris from the drain.

With reference to FIGS. 1 and 5, the sewer cleaning machine 1200 is shown in a second transport position. A user may tilt the frame 1210, lifting the housing 1214 off of a surface and allowing the wheels 1238 to transport the sewer cleaning machine 1200 along the surface. However, due to the weight, the sewer cleaning machine 1200 may be difficult to lift. Specifically, a user may have difficulty transporting the sewer cleaning machine 1200 on stairs. During transportation, the track 1226 may assist in lifting the sewer cleaning machine 1200 both up and down stairs. A user may first position the sewer cleaning device 1200 so that the track 1226 engages the stairs, for example as shown in FIG. 5. Once the track 1226 engages the stairs, the user can control the power supply 1222 to operate the second motor 1286. The second motor 1286 rotates the drive shaft 1258, thereby rotating the endless belts 1270, 1278. As the endless belts rotate 1270, 1278, the traction on the belts 1270, 1278 assist in pulling the sewer cleaning machine 1200 up the stairs. Meanwhile, the user can also assist by pulling on the handle 1230 of the sewer cleaning device 1200. To transport the sewer cleaning device 1200 down stairs, a user can control the second motor 1286 to rotate the track 1226 in the opposite direction.

With continued reference to FIG. 5, the center of gravity X_c of the sewer cleaning machine 1200 is positioned on the track 1226 while the sewer cleaning machine 1200 is at an angle relative to the ground (i.e., horizontal). In some embodiments, the angle of the sewer cleaning machine 1200 is between approximately 30 and 35 degrees with the horizontal. In other embodiments, the angle of the sewer cleaning machine 1200 is between approximately 32 and 33 degrees with the horizontal. In further embodiments, the angle of the sewer cleaning machine 1200 is 32.47 degrees with the horizontal, which correlates with the standard US code for the angle of stairs. Positioning the center of gravity X_c on the treads reduces the possibility of the sewer cleaning machine 1200 tipping if a user were to release the sewer cleaning machine 1200 while climbing or descending stairs. In some embodiments, when the sewer cleaning machine 1200 is tilted at an angle less than 35 degrees relative to the horizontal, the center of gravity X_c of the sewer cleaning machine may be above the tracks in a vertical direction. In a further embodiment, when the sewer cleaning machine 1200 is tilted at an angle of less than 33 degrees, the center of gravity X_c is above the treads in a vertical direction.

In some embodiments, the sewer cleaning machine 1200 includes a variable speed control that allows for stair climbing at a user selectable speed. For example, the sewer cleaning machine 1200 may travel up a set of stairs at a different speed than traveling down a set of stairs. In some embodiments, the sewer cleaning machine 1200 may travel up the set of stairs faster than the machine 1200 travels down the set of stairs. The user may adjust the speed of the track 1226 to travel up or down a set of stairs at a desired speed. In some embodiments, the controller 1284 is able to automatically control the speed of the track 1226 based on a direction of travel of the sewer cleaning machine 1200. For

example, the controller 1284 may determine whether the sewer cleaning machine 1200 is traveling up or down a set of stairs and automatically determine and adjust the desired speed of the track 1226. In some embodiments, the controller 1284 may determine when the track 1226 is rotating a first direction (e.g., clockwise) correlated with traveling up a set of stairs, and thereby increase the speed of the track 1226. Likewise, the controller 1284 may determine when the track 1226 is rotating in a second direction (e.g., counter-clockwise) correlated with traveling down a set of stairs, and thereby decrease the speed of the track 1226. In some embodiments, the user may control or may override the speed of the track 1226 designated by the controller 1284.

With reference to FIGS. 7-12, the track 1226 may include a belt tensioning assembly 1290. In the illustrated embodiment, the track 1226 includes two belt tensioning assemblies 1290, one of each endless belt 1270, 1278. In other embodiments, the track 1226 may include fewer or more belt tensioning assemblies 1290 (depending on the number of belts), or a single belt tensioning assembly may be used for multiple belts. The belt tensioning assemblies 1290 can increase or decrease the tensions of the first endless belt 1270 and the second endless belt 1278. For example, increasing the tension of the belts 1270, 1278 may reduce the possibility of the belts 1270, 1278 slipping during rotation. Increasing the tension of the belts 1270, 1278 may also reduce the possibility of the belts 1270, 1278 derailing from the pulley system. In addition, when the belts 1270, 1278 become worn and need to be replaced, the tension of the belts 1270, 1278 may be decreased in order to more easily remove the old belt and reattach a new belt. One of the belt tensioning assemblies 1290 is described in more detail below. The other belt tensioning assembly 1290 is substantially the same.

The illustrated belt tensioning assembly 1290 includes a bracket 1294, a mount 1298, and an adjustment member 1300. The bracket 1294 is movable relative to the mount 1298. The mount 1298 supports the adjustment member 1300, which helps to position the bracket 1294 relative to the mount 1298. The belt tensioning assembly 1290 is capable of adjusting the position of an idler roller, such as the first idler roller 1274 and the second idler roller 1282 relative to the first endless belt 1270 and the second endless belt 1278, respectively. As described herein, the belt tensioning assembly 1290 will be described with respect to the first idler roller 1274 and first endless belt 1270, however, it should be understood that the same construction can be applied to the second idler roller 1282 and the second endless belt 1278.

The bracket 1294 supports the first idler roller 1274. In particular, the bracket 1294 rotatably supports the first idler roller 1274 such that the first idler roller 1274 is rotatable relative to the bracket 1294. However, the bracket 1294 and the first idler roller 1274 translate together relative to the first endless belt 1270. As such, the bracket 1294 and the first idler roller 1274 can be translated in a first direction 1310 to increase the tension of the first endless belt 1270, and can be translated in a second direction 1312 to decrease the tension of the first endless belt 1270.

The mount 1298 is fixed to a portion of the frame 1210. The bracket 1294, and thus the first idler roller 1274, can translate relative to the mount 1298 in order to tension the first endless belt 1270. Specifically, the adjustment member 1300 is configured to adjust the position of the bracket 1294 and the first idler roller 1274 relative to the mount 1298. In the illustrated embodiment, the adjustment member 1300 is a threaded bolt 1304 and a nut 1308; however, in other embodiments, different types of adjustment members may

be used. The nut **1308** can be threaded along the length of the bolt **1304** to adjust the position of the bracket **1294**. For example, the bolt **1304** can be threaded in the first direction **1310** along the bolt **1304** to move the bracket **1294** in the first direction **1310** to increase tension in the first idler belt **1270**. Similarly, the bolt **1304** can be threaded in the second direction **1312** along the bolt **1304** to move the bracket **1294** in the second direction **1312** to decrease tension in the first idler belt **1270**. In addition, in the illustrated embodiment, the mount **1298** includes a pair of posts **1313** that extend through slots **1314** in the bracket **1294** to help align and guide the bracket **1294** as it moves relative to the mount **1298**.

In the illustrated embodiment, the nut **1308** can be accessed from both sides of the first endless belt **1270** through a window **1318** formed by the mount **1298**. This allows a user to use a wrench to thread the nut **1308** and adjust the tension of the first endless belt **1270**, thus reducing the amount of force input by the user.

FIG. **13** illustrates a transportable machine according to another embodiment. The transportable machine illustrated in FIG. **13** is a sewer cleaning machine **1400**. The sewer cleaning machine **1400** is similar to the sewer cleaning machine **1200** with like features being represented by like reference numerals. As will be understood, the features shown and described herein with respect to the other embodiments of a transportable machine may be used in combination with the features shown and described with respect to the sewer cleaning machine **1400**.

The sewer cleaning machine **1400** includes the frame **1410**, a handle **1430**, a base **1434**, and a track **1426**. The frame **1410** defines an elongated body extending between the handle **1430** and the base **1434**. Specifically, the handle **1430** is positioned on a top end of the frame **1410** to enable a user to grip and control movement of the sewer cleaning machine **1400**. The base **1434** is positioned on a bottom end of the frame **1410** and is configured to support a load, such as a machine or other load, that may be desirable to be able to transport. For example, the base **1434** may be configured to support many of the components of the sewer cleaning machine **1400**. In the illustrated embodiment, the base **1434** supports a drum housing **1414**, a motor housing **1418**, a power supply **1422**, and a track **1426** that is similar to the track **1226**. The track **1426** is supported by the frame **1410** to help facilitate transport of the sewer cleaning machine **1400**. In the illustrated embodiment, the track **1426** is positioned on a rear side of the frame **1410** between the handle **1430** and the base **1434**. Additionally, the illustrated sewer cleaning machine **1400** includes wheels rotatably coupled to the frame **1410** to assist in transport of the sewer cleaning machine **1400**.

Moving to FIG. **14**, the track **1426** includes a first extrusion **1424** coupled to the frame **1410** and a second extrusion **1428** coupled to the frame **1410** on the opposite side of the drive shaft **1458** as the first extrusion **1424**. The first and second extrusions **1424**, **1428** each respectively support the first and second drive rollers **1462**, **1466**, the first and second idler rollers **1474**, **1482**, the first and second endless belts **1470**, **1478** (FIG. **16**), and the belt tensioning assemblies **1490**. Each of the first and second extrusions **1424**, **1428** are coupled to frame **1410** with a bracket **1438** (FIG. **17**). The bracket **1438** couples an upper portion of the first and second extrusions **1424**, **1428** to the frame **1410**. The brackets **1438** are coupled to both the frame **1410** and first and second extrusions **1424**, **1428** using fasteners **1450** (FIG. **17**) that extend through fastener slots.

With reference to FIGS. **15** and **16**, both the first and second extrusions **1424**, **1428** include ribs **1436** that extend the length of the first and second extrusions **1424**, **1428** between the idler rollers **1474**, **1482** and the drive rollers **1462**, **1466**. In the illustrated embodiment, a first rib **1436** is on a front side of the first and second extrusions **1424**, **1428** facing the frame **1410** and a second rib **1436** is on a back side of the first and second extrusions **1424**, **1428** facing away from the frame **1410**. The ribs **1436** are generally placed in the center of the first and second extrusions **1424**, **1428**. The ribs **1436** each correspond to a recess **1440** on the inside of the endless belts **1470**, **1478**. The ribs **1436** hold the belts **1470**, **1478** in place while the belts **1470**, **1478** are being rotated. In the illustrated embodiment, each rib **1436** is a continuous projection extending along the length of the corresponding extrusion **1424**, **1428**. In other embodiments, the ribs **1436** may include discrete projections that only extend along portions of the length of the corresponding extrusion **1424**, **1428**. In other embodiments, the rib **1436** may only be on one side of the first and second extrusions **1424**, **1428**. For example, the rib **1436** may only be on a side facing away from the frame **1410** or only on a side facing the frame **1410**.

As seen in FIG. **17**, the track **1426** further includes a bracket **1438** coupled to the outside of each of the first and second extrusions **1424**, **1428**. The bracket **1438** includes a flange portion **1442** and a circular portion **1446**. The flange portion **1442** includes fastener holes for fasteners **1450** to extend through and couple the bracket **1438** to the respective extrusion **1424**, **1428**. The flange portion **1442** extends past the respective extrusion **1424**, **1428** towards the frame **1410**. The portion of the flange portion **1442** that extends past the first and second extrusions **1424**, **1428** keeps the belts **1470**, **1478** from slipping or jumping off the first and second extrusions **1424**, **1428** during operation of the track **1426**. Each circular portion **1446** includes an opening **1454** adjacent the respective drive rollers **1462**, **1466** to protect the drive rollers **1462**, **1466** during operation of the track **1426**.

With continued reference to FIG. **17**, an inner support **1460**, or guide block, is supported on the inside of each of the first and second extrusions **1424**, **1428**. The fasteners **1450** used to couple the brackets **1438** to the first and second extrusions **1424**, **1428** are also used to support the inner supports **1460** within the first and second extrusions **1424**, **1428**. Each inner support **1460** includes an outer surface **1464** that extends from within the first and second extrusions **1424**, **1428**. The outer surface **1464** is generally aligned with or flush with an adjacent outer surface of the corresponding extrusion **1424**, **1428**. The outer surface **1464** assists the belts **1470**, **1478** in transitioning from the first and second extrusions **1424**, **1428** to the cleats of the drive rollers **1462**, **1466**. An inner surface of the inner support **1460** is curved or arcuate to match the curvature of the corresponding drive roller **1462**, **1466**. As such, the inner support **1460** can be positioned close to, but not interfere with the corresponding driver roller **1462**, **1466**.

With reference to FIGS. **13**, **14**, and **18**, the sewer cleaning machine **1400** further includes a support system **1406**. The support system **1406** helps stabilize the sewer cleaning machine **1400** when in the upright position. The support system **1406** includes a first stationary kickstand **1408**, a second stationary kickstand **1412**, and a tilt block **1416** (FIG. **14**). The first and second kickstands **1408**, **1412** and the tilt block **1416** are generally supported beneath the frame **1410** when the sewer cleaning machine **1400** is in the upright position. In some embodiments, the support system

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1406 may be considered part of the frame 1410 of the sewer cleaning machine 1400. The first and second kickstands 1408, 1412 are generally tubular and are fixed relative to the frame 1410. The first kickstand 1408 is generally positioned in front of the drum housing 1414 underneath the feed device 1250. The second kickstand 1412 is positioned behind the drum housing 1414. Both the first and second kickstands 1408, 1412 include rubber ends 1420 to inhibit the sewer cleaning machine 1400 from moving when in the upright position. The tilt block 1416 is positioned behind the second kickstand 1412 relative to the drum housing 1414. The tilt block 1416 may be solid to add weight to the support system 1406. The tilt block 1416 inhibits the sewer cleaning machine 1400 from tipping backwards when a cable is removed from the drum housing 1414 and the sewer cleaning machine 1400 is back heavy.

In some embodiments, the sewer cleaning machine 1400 may include a pop-up wheel or bracket. The pop-up wheel or bracket may be activated to inhibit the sewer cleaning machine from falling down a flight of stairs if a user were to accidentally lose control or hold of the sewer cleaning machine 1400 while climbing stairs. Additionally, the pop-up wheel or bracket may be extended in between climbing steps of a flight of stairs to help support the sewer cleaning machine and inhibit the sewer cleaning machine from tipping backwards or forward. The pop-up wheel may be activated either manually or with a spring release to extend from under the frame 1410 of the sewer cleaning machine 1400. In this way, when there is a pause in stair climbing the frame 1201 would rest on the pop-up bar or bracket that is supported by the previous step.

FIGS. 19-24 illustrate a transportable machine according to another embodiment of the disclosure. The transportable machine illustrated in FIGS. 19-24 is a dolly or hand truck 2010. The hand truck 2010 includes structure similar to the sewer cleaning machines 1200, 1400 of the previous embodiments with like features being represented by like reference numerals. As will be understood, the features shown and described herein with respect to the other embodiments of a transportable machine may be used in combination with the features shown and described with respect to the hand truck 2010.

The hand truck 2010 includes a frame 2210, a handle 2230, a base 2234, and a track 2226. The frame 2210 defines an elongated body extending between the handle 1430 and the base 2234. Specifically, the handle 1430 is positioned on a top end of the frame 2210 to enable a user to grip and control movement of the hand truck 2010. The base 2234 is positioned on a bottom end of the frame 2210 and is configured to support a load, such as a machine or other load, that may be desirable to be able to transport. For example, the base 2234 may be configured to support a machine, tools, cargo, building materials, or other heavy items that may be difficult to transport. In the illustrated embodiment, the base 2234 includes a loading plate member 2040. The loading plate member 2040 and the frame 2210 cooperatively define a space 2044 configured to receive and support a load (e.g., boxes, furniture, appliances, power tools, other goods, and the like) on the hand truck 2010. The track 2226 is supported by the frame 2210 to help facilitate transport of the hand truck 2010. In the illustrated embodiment, the track 2226 is positioned on a rear side of the frame 2210 between the handle 2230 and the base 2234. Additionally, the illustrated hand truck 2010 includes wheels 2238 rotatably coupled to the frame 2210 to assist in transport of the hand truck 2010.

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The track 2226 is configured to engage a surface, such as stairs or a ramp to help a user maneuver the hand truck 2010. In the illustrated embodiment, the track 2226 is positioned on the opposite side of the frame 2210 as the space 2044. For example, the space 2044 is defined on a front side of the frame 2210, and the track 2226 is positioned on a back side of the frame 2210. The track 2226 includes a substantially horizontal drive shaft having a first endless belt 2270 that extends around a first drive roller and a first idler roller, and a second endless belt 2278 that extends around the second drive roller and a second idler roller. The endless belts 2270, 2278 extend substantially vertically along a length of the frame 2210. A motor 2286 (FIG. 20) is supported by the frame 2210 and is coupled to the drive shaft of the track 2226. The motor 2286 is operable to rotate the drive shaft and thus the drive rollers to facilitate rotation of the endless belts 2270, 2278. The motor 2286 is powered by a power supply 2222 that is supported on the frame 2210. In some embodiments, the endless belts 2270, 2278 include traction elements that assists in gripping a surface, ledge, or other object.

A user may tilt the frame 2210, lifting the loading plate member 2040 off of a surface and allowing the wheels 2238 to transport the hand truck 2010 along the surface. However, due to the weight, the hand truck 2010 may be difficult to lift. Specifically, a user may have difficulty transporting the hand truck 2010 on stairs or into a truck bed. During transportation, the track 2226 may assist in lifting the hand truck 2010 both up and down stairs. A user may first position the hand truck 2010 so that the track 2226 engages the stairs, for example as shown in FIG. 24. Once the track 2226 engages the stairs, the user can control the power supply 2222 to operate the motor 2286. The motor 2286 rotates the endless belts 2270, 2278. As the endless belts rotate 2270, 2278, the traction on the belts 2270, 2278 assist in pulling the hand truck 2010 up the stairs. Meanwhile, the user can also assist by pulling on the handle 2230 of the hand truck 2010. To transport the hand truck 2010 down stairs, a user can control the motor 2286 to rotate the track 2226 in the opposite direction.

In some embodiments, the track 2226 can articulate or expand and retract relative to the frame 2210 to better climb stairs. For example, FIG. 19 illustrates the hand truck 2010 in a first (stored) position. In this position, the loading plate member 2040 of the base 2234 of the hand truck 2010 is supported on a surface to facilitate supporting of the load. FIGS. 20 and 24 illustrate the hand truck 2010 in a second, extended (operational) position. In the second position, the track 2226 is moved or pivoted away from the frame 2210. In the illustrated embodiment, the track 2226 includes an axle positioned proximate the base 2234, and the track 2226 is movable (e.g., pivotable) between the first position and the second position about the axle. The illustrated hand truck 2010 further includes a lock mechanism 2048 for maintaining the track 2226 in the second position. The illustrated lock mechanism 2048 includes a sub frame 2052 movably coupled to the frame 2210, and a latch 2056 (i.e., handle latch) selectively coupled to the sub frame 2052. The latch 2056 may be operated by a user using one hand. In particular, when the hand truck 2010 is in the second, extended position, the wheels 2238 are tilted away from the stairs such that the only the track 2226 is configured to contact the stairs.

With reference to FIGS. 21-23, the hand truck 2010 further includes a lifting system 2060 to facilitate tilting of the hand truck 2010 toward or away from a surface. In the illustrated embodiment, as shown in FIGS. 21-23, the lifting

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system **2060** includes an air bag member **2064** supported by the base **2234**. More specifically, the illustrated air bag member **2064** is positioned between the loading plate member **2040** and a lifting plate member **2068** of the base **2234**. The lifting system **2060** further includes a pump **2072** and a foot pedal member **2076** supported by the frame **2210** and track **2226**. The pump **2072** is electrically coupled to the power supply **2222**. The illustrated pump **2072** and foot pedal **2076** are positioned proximate the back side of the frame **2210**. The air bag member **2064** is configured to be inflated such that the loading plate member **2040** moves or pivots relative to the lifting plate member **2068**. The inflation of the air bag member **2064** adjusts the hand truck **2010** from a vertical or loading state to a tilted state (FIG. **23**). A release member **2080** is provided proximate the handle **2230** for releasing or deflating the air bag member **2064** from the tilted state to the loading state. The release member **2080** includes an actuator (e.g., lever) operable by a user. In other embodiments, instead of the air bag member **2064**, the lifting system **2060** includes a powered actuator such as a gas cylinder, a piston, or other mechanical means, to pivot the loading plate member **2040** relative to the lifting plate member **2068**. Still further, in other embodiments, the lifting plate member **2068** may include a wheel coupled thereto to facilitate guiding of the hand truck **2010** on a flat surface. The lifting system **2060** is configured to facilitate tilting of the hand truck **2010** toward the track **2226**, thereby reducing the need to manual tilt the hand truck **2010** toward the track **2226**.

The handle **2230** includes a control system having a plurality of switches **2083** positioned on the handle **2230** and a control panel **2084** positioned proximate a top side of the hand truck **2010**. The control system includes a controller electrically connected to the motor **2286** for controlling operation of the hand truck **2010**. In particular, the plurality of switches **2083** on the handle **2230** includes an actuator (e.g., trigger) for controlling operation of the track **2226**. In the illustrated embodiment, the switches **2083** include a first switch for turning on and off the motor **2286**. Still further, in other embodiments, the handle **2230** may also contain controls for additional functions of the stair climbing hand truck **2010** (e.g., variable speed). The illustrated control panel **2084** is positioned on the top side of the stair climbing hand truck **2010** to facilitate access of the user to the control panel **2084** for changing settings, features and other functions of the hand truck **2010**.

FIGS. **25-26** illustrate a dolly or hand truck **2110** according to another embodiment of the disclosure. The hand truck **2110** includes structure similar to the hand truck **2010** with like features being represented by like reference numerals. As will be understood, the features shown and described herein with respect to the other embodiments of a transportable machine may be used in combination with the features shown and described with respect to the hand truck **2110**.

In this embodiment, the handle **2230'** is movable between a first, angled position (FIG. **25**) and a second, extended position (FIG. **26**). In the illustrated embodiment, the handle **2230'** is pivotable about a pin **2088**. The handle **2230'** further includes an adjustment mechanism for adjusting the handle **2230'** between the first position and the second position. The illustrated adjustment mechanism includes an actuator **2092** (e.g., pad) operatively coupled to the pin **2088**. A user may actuate the actuator **2092** (e.g., push) for disengaging the pin **2088** from a detent mechanism, thereby adjusting the handle **2230'** between the first and second positions. The first position may facilitate movement of the hand truck **2110** on

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a flat surface by a user. The second position may facilitate movement of the hand truck **2110** on stairs by a user.

FIGS. **27-28** illustrates a dolly or hand truck **2310** according to another embodiment of the disclosure with like features being represented by like reference numerals. As will be understood, the features shown and described herein with respect to the other embodiments of a transportable machine may be used in combination with the features shown and described with respect to the hand truck **2310**.

The hand truck **2310** includes structure similar to the hand truck **2010** with like features being represented by like reference numerals. In this embodiment, the hand truck **2310** includes a securing mechanism **2096** to facilitate securing of a load **3000** received in the space **2044** defined by the hand truck **2310**. In the illustrated embodiment, the hand truck **2310** includes a plurality of adjustable straps **2096A-C**. In some embodiments, the adjustable straps **2096A-C** are ratchet straps. Additionally or alternatively, the adjustable straps **2096A-C** may be retractable straps. Each of the straps **2096A-C** is supported by the frame **2210**. In particular, each strap **2096A-C** extends from a first frame member **2210A** of the frame **2210** to a second frame member **2210B** in a lateral direction relative to the space **2044** configured to receive the load **3000**. Each strap **2096A-C** is adjustable to individually lengthen or shorten each strap **2096A-C**. In some embodiments, one or more of the straps **2096A-C** is automatically retractable. In the illustrated embodiment, the securing mechanism **2096** includes three straps **2096A-C**. In other embodiments, the hand truck **2310** may include one or more straps. As shown in FIG. **28**, the ratchet mechanism of each adjustable strap **2096A-C** is secured to a support member **3004** (e.g., loop) positioned on the first frame member **2210A**.

Although the disclosure is described with reference to discrete embodiments, variations of the sewer cleaning machines and hand trucks exist within the spirit and scope of the disclosure. The features shown and described herein with respect to one of the embodiments of a transportable machine may be used in combination with any of the features shown and described with respect to another embodiment of a transportable machine. For example, the following features may be used in any combination with one another, including but not limited to, the belt tensioning assembly (e.g., shown in FIGS. **7-12**), the stands (e.g., shown in FIG. **18**), the movable track assembly (e.g., shown in FIG. **20**), the lifting assembly (e.g., shown in FIG. **21-23**), or the movable handle (e.g., shown in FIGS. **25-26**). Additionally, any of the aforementioned features may be used in combination with the various control features, including but not limited to, the lock out system, which prevents a machine motor from being operated simultaneously with the track motor, a power monitoring system, which warns a user or shuts down the track motor when the power is insufficient to climb a full set of stairs, or a variable speed control system that operates the track at different speeds depending on whether the transportable machine is climbing up or down a set of stairs.

What is claimed is:

1. A transportable machine, comprising:
 - an elongated frame having a top end and a bottom end; a handle disposed on the top end of the frame;
 - a base disposed on the bottom end of the frame, the base and the frame together defining a space configured to receive and support a load;
 - a wheel disposed on the bottom end of the frame;

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- a track coupled to the frame, the track including
 a first roller,
 a second roller,
 an endless belt configured to rotate about the first and
 second rollers, and
 a belt tensioning assembly configured to tension the
 endless belt, the belt tensioning assembly including
 a bracket supporting the first roller, a mount coupled
 to the frame, and a single adjustment member acces-
 sible on each side of the endless belt and configured
 to adjust the position of the bracket relative to the
 mount to adjust a tension of the endless belt; and
 a motor coupled to the second roller, the motor operable
 to drive the endless belt.
2. The transportable machine of claim 1, wherein the
 endless belt includes a recess extending along an inside
 surface, the recess receiving a rib on the frame to align the
 endless belt relative to the frame.
3. The transportable machine of claim 1, wherein the track
 further includes a third roller, a fourth roller, and a second
 endless belt configured to rotate about the third and fourth
 rollers.
4. The transportable machine of claim 1, further compris-
 ing a sewer cleaning machine supported by the base, the
 sewer cleaning machine including a drum, a cable posi-
 tioned at least partially within the drum, and a second motor
 driving rotation of the drum.
5. The transportable machine of claim 1, wherein the
 handle is movable relative to the elongated frame between
 an angled position, in which the handle extends generally
 downward from the top end of the frame, and an extended
 position, in which the handle extends generally upward from
 the top end of the frame.
6. The transportable machine of claim 1, wherein the track
 is movable relative to the elongated frame between a stored
 position and an extended position, wherein the track is
 positioned farther away from the frame when in the
 extended position than in the stored position.
7. The transportable machine of claim 1, wherein the
 single adjustment member is positioned between the first
 roller and the second roller.

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8. A transportable machine, comprising:
 an elongated frame having a top end and a bottom end;
 a handle disposed on the top end of the frame;
 a base disposed on the bottom end of the frame, the base
 and the frame together defining a space configured to
 receive and support a load;
 a wheel disposed on the bottom end of the frame;
 a track coupled to the frame, the track including a first
 roller, a second roller, a frame member extending
 between the first and second rollers, and an endless belt
 configured to rotate about the first and second rollers,
 the endless belt having a recess formed on an inside
 surface of the endless belt, the recess receiving a rib on
 the frame member to align the endless belt relative to
 the frame member; and
 a motor coupled to the second roller, the motor operable
 to drive the endless belt.
9. The transportable machine of claim 8, wherein the track
 further includes a bracket positioned adjacent the endless
 belt and configured to inhibit the endless belt from slipping
 off of the frame.
10. The transportable machine of claim 8, wherein the
 track further includes a third roller, a fourth roller, and a
 second endless belt configured to rotate about the third and
 fourth rollers.
11. The transportable machine of claim 8, further com-
 prising a belt tensioning assembly configured to tension the
 endless belt.
12. The transportable machine of claim 8, further com-
 prising a sewer cleaning machine supported by the base, the
 sewer cleaning machine including a drum, a cable posi-
 tioned at least partially within the drum, and a second motor
 driving rotation of the drum.
13. The transportable machine of claim 12, wherein
 operation of one of the first motor and the second motor
 prevents operation of the other of the first motor and the
 second motor.
14. The transportable machine of claim 8, wherein the
 recess also receives a rib on the second roller to align the
 endless belt relative to the frame member.

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